RIGOROUS INTEGER FACTORISATION IN EXPECTED TIME $L_N[1/2, c]$ FOR EVERY $c > 1/\sqrt{2}$

ABSTRACT. We describe a Las Vegas algorithm which, given any integer $N \geq 2$, outputs the complete prime factorisation of N, and we give a proof that for every $c > 1/\sqrt{2}$ its parameters can be fixed so that its expected running time is $O(L_N[1/2, c])$, where $L_N[1/2, c] = \exp\big(c\sqrt{\log N \log\log N}\big)$. The proof uses no unproved hypothesis. The previous best rigorous bound for factoring every integer is $L_N[1/2, 1 + o(1)]$, due to Lenstra and Pomerance (1992).

For each fixed choice of parameters (an odd degree d, a number k of large primes, a base exponent μ and a slack η_0) the expected time is $L_N[1/2, (1+\theta)/\sqrt{2}+o(1)]$, where $\theta = \theta(d, k, \mu, \eta_0)$ is an explicit exponent. For $d = 9$ and $k = 13$ this is below 1 ($c \approx 0.9972$). The infimum over fixed parameters at $d = 9$ is $1.3/\sqrt{2} \approx 0.9192$, and over all d it is $1/\sqrt{2}$. These are infima: we do not prove that a single algorithm of fixed degree attains $(1 + 3/(d + 1))/\sqrt{2} + o(1)$.

The algorithm combines four ingredients. The first is a "Dixon descent": a uniformly random square $r^2 \bmod N$ is written as a ratio u/v of two integers of size $\sqrt{N}$, and each prime of uv is descended by a special-q relation in a randomised number field sieve of fixed degree. Because the root r is never used, every dependency is fruitful with probability $1 - 2^{1-\omega(N)}$. Second, the halves u, v are completely factored with the Lenstra–Pomerance algorithm at cost $L_{\sqrt{N}}[1/2, 1] = L_N[1/2, 1/\sqrt{2}]$, rather than being combined. Third, the supply of relations is reduced to a single averaged estimate, which we prove by the Paley–Zygmund inequality from a first moment (a restricted Bombieri–Vinogradov theorem for the moduli $\ell \cdot$ (smooth), in the spirit of Lee–Venkatesan) and a second moment. The second moment is made tractable by randomising the rational side in two dimensions and by a random prime multiplier w of the resultant, $F(g_0, g_1) = Nw$, which spreads a forced residue class. It is then proved with divisor switching and the large sieve. Fourth, the number field sieve post-processing of Buhler–Lenstra–Pomerance and Lee–Venkatesan is carried out rigorously at fixed degree, and batch smoothness detection removes any dependence on the hyperelliptic curve method.

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1. Introduction

1.1. **Background.** Write $L_N[\alpha, c] = \exp\big(c(\log N)^{\alpha}(\log\log N)^{1-\alpha}\big)$ and $L = L_N[1/2, 1]$. The fastest factoring algorithms in practice, the quadratic and number field sieves, run in heuristic

time $L_N[1/2, 1 + o(1)]$ and $L_N[1/3, (64/9)^{1/3} + o(1)]$ [41, 31, 6]. Their running times are not proved: they rest on assumptions that certain polynomial values behave like random integers with respect to smoothness, and that the resulting congruences of squares are not all trivial.

Rigorous analysis has a separate history.

- Dixon's random-squares method [13] was the first factoring algorithm with a proved subexponential running time, $L_N[1/2, O(1)]$. Its correctness argument is simple and is the starting point of this paper. Draw r uniformly, look for relations $r^2 \equiv \prod p^{e_p} \pmod N$ over a factor base, and combine them into $X^2 \equiv Y^2$. Since Y is computed without looking at the roots r, the class of X/Y in the group of square roots of unity is uniformly distributed, and $\gcd(X - Y, N)$ is a proper factor with probability at least $1/2$. This is the same fibre symmetry that underlies Rabin's reduction from computing square roots to factoring [44].
- Pomerance analysed and refined these methods [41], and proved $L_N[1/2, \sqrt{2} + o(1)]$ for random squares with a rigorous elliptic curve smoothness test [42]. Vallée's sampler of small squares gave $L_N[1/2, \sqrt{4/3} + o(1)]$ [52].
- Class-group methods reached exponent 1: Schnorr and Lenstra [46] under hypotheses, and Seysen [47] and A. K. Lenstra [30] under the generalised Riemann hypothesis. Lenstra and Pomerance [33] removed every hypothesis. They proved that the complete prime factorisation of every N can be found in expected time $L_N[1/2, 1 + o(1)]$. That bound has stood since 1992. It is still described as the best rigorous probabilistic bound, for instance by Pomerance [43] and Harvey [20].
- Lee and Venkatesan [29] analysed rigorously an explicitly randomised number field sieve. They proved that it finds congruences of squares in expected time $L_N[1/3, (64/9)^{1/3} + o(1)]$. Whether these congruences yield a factor is left to a conjecture (their Conjecture 7.1), and their factoring conclusion is stated for $N = pq$ with $p \equiv q \equiv 3 \pmod 4$. Their techniques for the first moment of smooth values (a randomisation that makes algebraic values nearly uniform on long progressions, and a restricted Bombieri–Vinogradov estimate for smooth moduli built on Harper's work [18]) and for the post-processing (random characters, Stark–Landau) are the models for ours.

There is also a folklore heuristic that $L[1/2, 1]$ is a natural barrier for methods that combine random-like auxiliary integers of size at least $N^{1/2}$ into a square. Croot, Granville, Pemantle and Tetali [10] and Balister, Bollobás and Morris [2] made it sharp for random integers. We found no theorem showing that $c = 1$ is a barrier for rigorous factoring itself.

1.2. **The result.** All our parameters are fixed: an odd degree $d \geq 3$, an integer $k > (d + 1)/4$, a rational base exponent μ and a slack $\eta_0 > 0$. Put $\nu = 1/(4k)$ and

$$(1)\quad h_*(\mu) = \max\bigl(2\nu,\ \mu-(d-2)\nu,\ 1-d\mu+\nu\bigr), \qquad \theta = \theta(d, k, \mu, \eta_0) = 2\mu+h_*(\mu)+\eta_0+(d-1)\nu.$$

The parameters are *admissible* if $\nu + \eta_0 \leq \mu < 1$ and $\theta < 1$.

Theorem 1.1 (main theorem, fixed parameters)**.** *For every admissible (d, k, μ, η_0) there is a Las Vegas algorithm which, given any integer $N \geq 2$, outputs the complete prime factorisation of N. Its expected running time is*

$$L_N\bigl[1/2,\ (1 + \theta)/\sqrt{2} + o(1)\bigr] \qquad (N \to \infty),$$

where the $o(1)$ depends only on (d, k, μ, η_0).

Minimising θ over μ (Lemma 3.2) gives, at $\eta_0 \downarrow 0$,

$$\theta_{\min}(d,k) = \begin{cases} \dfrac{2}{d} + \dfrac{d^2+d-2}{d}\,\nu, & k < (d^2+1)/4, \\ \dfrac{3}{d+1} + \dfrac{4d-2}{d+1}\,\nu, & k \geq (d^2+1)/4, \end{cases} \qquad \inf_{k,\eta_0} \frac{1+\theta}{\sqrt{2}} = \frac{1+3/(d+1)}{\sqrt{2}}.$$

Letting d grow gives the uniform statement.

Corollary 1.2. *For every $c > 1/\sqrt{2}$ there is a Las Vegas algorithm which, given any integer $N \geq 2$, outputs its complete prime factorisation in expected time $O(L_N[1/2, c])$.*

Proof. Choose d with $(1+3/(d+1))/\sqrt{2} < c$, then k large and η_0 small with $(1+\theta)/\sqrt{2} < c$, and apply Theorem 1.1. Finitely many N cost $O(1)$. □

Table 1 lists exact values. The constant is below 1 if and only if $\theta < \sqrt{2} - 1 = 0.41421\ldots$.

d	least k	θ there ($\eta_0 \downarrow 0$)	c there	need $\eta_0 <$	c at $k = 50$	$k = 1000$	$\inf_{k,\eta_0} c$
7	21	139/336	0.99963	0.00052	0.98376	0.97285	0.97227
9	13	16/39	0.99720	0.00396	0.93126	0.91984	0.91924
11	13	9/22	0.99638	0.00512	0.89626	0.88450	0.88388
13	14	73/182	0.99073	0.01311	0.87126	0.85926	0.85863
19	17	257/646	0.98842	0.01638	0.85188	0.81383	0.81317
$d \to \infty$							$1/\sqrt{2} = 0.70711$

TABLE 1. The constant $c = (1+\theta)/\sqrt{2}$ of Theorem 1.1 at the optimal μ. "Least k" is the least k with $c < 1$ for all sufficiently small $\eta_0 > 0$. At $d = 9$, $k = 13$: $\mu = 17/156$, $h_* = 1/26$, and $(1 + 16/39)^2 = 3025/1521 < 2 = 3042/1521$; at $k = 12$ the minimum is $\theta = 23/54 > \sqrt{2} - 1$. For $d \leq 5$ the constant is never below 1 ($d = 5$: infimum 1.0607). Computed in exact rational arithmetic by `src/thmA/params.py`; independently re-derived in two referee rounds.

Remark 1.3 (what is and is not claimed)**.** (a) *Infimum, not $o(1)$.* At fixed d the value $(1+3/(d+1))/\sqrt{2}$ is an infimum over fixed (k, η_0). It is not attained, and we do not claim $L_N[1/2, (1+3/(d+1))/\sqrt{2} + o(1)]$ for any fixed d. That would require letting $k = k(N) \to \infty$. Our constants depend on k through factors $(\log N)^{O(k)}$, $\binom{2k}{k}$ and thresholds $N_0(k)$, and we have not carried out such a diagonalisation. At $d = 9$ the best single algorithm we analyse has $c \approx 0.9972 + \eta_0/\sqrt{2}$.

(b) *Complete factorisation.* The algorithm outputs the full factorisation, with every prime certified (AKS [1]). Prime powers, perfect powers and small prime factors are handled before the main splitting routine. The splitting routine is applied at most $\log_2 N$ times (Section 13).

(c) *No hypotheses, no exceptional set.* Nothing is assumed beyond published theorems. In particular no Riemann hypothesis, no hyperelliptic curve method (whose published proof trail is incomplete), and no exceptional set of N. Every input is a published, numbered statement. One of them, Drappeau's Lemme 5, has a published proof that points to computations in Harper's unpublished preprint (Appendix A). The threshold beyond which the time bound holds involves Stark's, Landau's and Lagarias–Odlyzko's effective constants, so it is effective but enormous; the algorithm itself uses no analytic constant (Remark A.17). The bound holds for every N, not on average.

(d) *Status.* The proof is new and has not been peer reviewed. It is long and rests on several delicate interfaces, so it should be treated with caution until experts have checked it.

1.3. **Comparison with Lenstra–Pomerance.**

	Lenstra–Pomerance 1992 [33]	this paper
output	complete factorisation, Las Vegas	complete factorisation, Las Vegas
hypotheses	none	none
expected time	$L_N[1/2, 1 + o(1)]$	$L_N[1/2, (1+\theta)/\sqrt{2} + o(1)]$, $\theta < 1$ fixed
best constant	1	any $c > 1/\sqrt{2}$ (Corollary 1.2)
relations	class-group relations with multipliers	Dixon descent in a randomised NFS
smoothness tests	ECM on "recognisable" smooth numbers	batch smooth parts (no ECM/HECM)
use of the other	—	LP factors the two $\sqrt{N}$-size halves

We use the Lenstra–Pomerance algorithm as a subroutine. It factors integers of size $\sqrt{2N}$ in time $L_{\sqrt{N}}[1/2, 1 + o(1)] = L_N[1/2, 1/\sqrt{2} + o(1)]$, and that is exactly where the constant $1/\sqrt{2}$ comes from. Any improvement of the rigorous constant for factoring $\sqrt{N}$-size integers would improve ours (Section 15).

1.4. **The idea in one page.**

Fruitfulness without characters. Take r uniform in $(\mathbb{Z}/N)^\times$ and $R = r^2$. Suppose we can express R multiplicatively over some factor base, in a way that depends on R and on independent coins but never on r. Then the exponent vectors of many such R can be combined by linear algebra, and the resulting congruence $\mathsf{X}^2 \equiv \mathsf{Y}^2$ has X/Y uniform in $\boldsymbol{\mu}_2 = \{\epsilon : \epsilon^2 = 1\} \cong \{\pm 1\}^{\omega(N)}$. So it is fruitful with probability exactly $1 - 2^{1-\omega(N)}$, whatever the factor base is (Theorem 6.1).

Dixon's method expresses R itself as smooth, which is expensive. We *descend* it instead, as the individual-logarithm phase of number field sieve discrete-log algorithms does [17, 25, 9]. Write $R \equiv u/v \pmod N$ with $|u|, |v| \le \sqrt{2N}$, by Gauss reduction of a lattice of determinant N. Then express each prime $\ell \mid uv$ through a special-q relation of a number field sieve.

The shape test. We accept R only if $|u|$ and $|v|$ are products of at most k distinct primes, all in one dyadic interval J around $N^{1/(2k)}$. Deciding this requires complete factorisation of $\sqrt{N}$-size integers whose prime factors are $N^{\Theta(1)}$. We pay for it with the Lenstra–Pomerance algorithm, at $L_N[1/2, 1/\sqrt{2}]$ per trial. With k fixed, a "good" field (one in which a fraction $\delta \ge L^{-o(1)}$ of the primes of J can be descended) accepts a fraction $\ge (\delta/\log N)^{2k} 2^{-\omega(N)} = L^{-o(1)}$ of all R. This follows from a Dixon–Pomerance-type counting lemma in ratio form (Theorems 5.1 and 5.2), which holds for an arbitrary set of admissible primes.

One averaged estimate suffices. Theorem 7.2 shows that complete descents in one field follow from a single averaged statement: $\Pr_{A,\ell}[\ell \text{ descends in field } A] \ge L^{-o(1)}$, where A is a random field and ℓ a random prime of J. No joint event across the k targets of a split is needed, because each field only uses splits inside its own set of descendable primes.

The field and the moments. The field is $F_A = \hat{F} + G \cdot R_c$, a binary form of degree d, where:

- $G(x, y) = g_1 x - g_0 y$ with (g_0, g_1) random of size $M = N^\mu$;
- $\hat{F}$ is the base-(g_0, g_1) expansion of Nw for a random prime w slightly larger than MN^ν;
- R_c has random coefficients of size $H = N^h$.

A prime $\ell \in J$ descends if some primitive $P = (a, b)$ in the special-q lattice $\{\ell \mid G(P)\}$, in a box of side $V \approx N^\nu$, has $G(P)/\ell$ and $F_A(P)$ both B-smooth with $B = L^{\theta/\sqrt{2}}$. The two smooth values have sizes $N^{\mu-\nu}$ and $N^{\mu+h+d\nu}$, with total exponent θ. The Paley–Zygmund inequality reduces the averaged statement to the following two moment bounds.

- A *first moment* (Theorem 8.7). The algebraic values lie in a progression to the modulus $|G(P)|$, which is a multiple of ℓ. The randomisation of R_c makes them nearly uniform on long progressions, as in Lee–Venkatesan. A restricted Bombieri–Vinogradov estimate for the moduli $\ell \cdot$ (smooth) (Lemma 8.4) shows that bad moduli are rare.
- A *second moment* (Theorem 9.7). Pairs of candidates in one lattice are coupled through a kernel lattice of index $|\det(P, P')|$ (Lemma 9.2). The forced residue $F_A(P) \equiv N w\, b^d g_1^{-d} \pmod{G(P)}$ would couple them through the rational data. The random prime w spreads it (Lemma 9.3, Brun–Titchmarsh). What is left is an all-residue second moment of smooth numbers to the moduli $\ell^2 nq$. We prove it at the precision we need by divisor switching and the large sieve, averaged over ℓ (Theorem 9.5).

Cost. About B rows are needed. Each costs $L^{1/\sqrt{2}}$ for the shape test and $S^2 \approx L^{\theta/(2a)}$ for the special-q searches, where $B = L^a$. Linear algebra costs B^2. The total is $L^{\max(2a,\, a+1/\sqrt{2},\, a+\theta/(2a))}$, minimised at $a = \theta/\sqrt{2}$, which gives $(1+\theta)/\sqrt{2}$. The folklore $L[1/2, 1]$ model does not bind. The two $\sqrt{N}$-size halves are factored, not combined. The relations that enter the linear algebra are descent norms of total size N^θ with $\theta < \sqrt{2} - 1 < 1/2$.

1.5. **Prior art and attributions.** Most components are known. We list them with the precise debt; the combination is what we did not find.

- *Fibre symmetry and fruitfulness.* Rabin [44] and Dixon [13], with Pomerance's analyses [41, 42]. Theorem 6.1 is this argument applied to a descent; we regard it as folklore-level.
- *Descent.* Rational reconstruction of a target as u/v, followed by special-q descent of its large primes, is the initial step of number field sieve individual-logarithm algorithms [17, 25, 9]. These analyses are heuristic. Stange [48] studies index calculus modulo a composite n with random exponents and multiplicative relations, with an NFS variant, under her Hypothesis 3.1. She is the closest conceptual precedent for using hidden random elements as targets.
- *Split counting.* Theorems 5.1 and 5.2 adapt the Dixon–Pomerance random-squares counting [13, 41] to the ratio form $u \equiv Rv$.
- *Non-monic linear side and post-processing.* The linear form $g_1 x - g_0 y$ with a base-(g_0, g_1) expansion is the base-(m_1, m_2) variant of Buhler, Lenstra and Pomerance [6, §§3,12]; see also Murphy [38, §2.3.2] and Kleinjung [27]. Our post-processing follows BLP's obstruction-group analysis, their proposal of random quadratic characters, their p-adic square root and their parity coordinate for the leading coefficient [6, §§6,8,9,12], the last after Weinberger–Rothschild [54]. It also follows Lee–Venkatesan's rigorous treatment of characters through Stark and Landau [29, §6]. What we add is the fixed-degree specialisation with explicit bounds (Section 10).
- *Randomised NFS and the first moment.* The randomisation $f = \hat{f} + (x - m)R$ and the use of a restricted Bombieri–Vinogradov estimate over smooth moduli are due to Lee–Venkatesan [29, §§5,8], building on Harper's Bombieri–Vinogradov and Barban–Davenport–Halberstam theorems for smooth numbers [18]. Our Lemma 8.4 is the analogue for the moduli $\ell\cdot$(smooth). Its proof (Appendix A) is self-contained apart from one published lemma of Drappeau [14, Lemme 5] (also Pascadi [39, Lemma 3.2]) for characters of small conductor, and the classical large sieve [24, Thm. 7.13]. Our one-point smoothing Lemma 8.1 replaces Lee–Venkatesan's Lemmas 5.13–5.20.
- *Second moment.* Lemma 9.4 is a progression form of the standard flexible factorisation of smooth numbers [15, 21] (cf. [19, Smooth Numbers Result 3]). Theorem 9.5 follows the classical large-sieve route to Barban–Davenport–Halberstam upper bounds [16, 34], with a Hooley-type rearrangement [23]. We found no statement of it.

- *The multiplier.* Multipliers are an old device. Knuth–Schroeppel multipliers kN appear in the continued fraction and quadratic sieve methods [37]; Schnorr–Lenstra [46] and Lenstra–Pomerance [33] use multipliers of the discriminant (LP: "the main device ... is the use of multipliers"); and kN is used in NFS polynomial selection [38]. Our use differs. The multiplier w is a large random prime of the resultant, of the size of the rational modulus, chosen to spread a forced residue in a rigorous second moment. We found no earlier instance of this.
- *Batch smoothness.* Bernstein's smooth-parts algorithm [4, 5]. Bernstein proposed it as a replacement for ECM in rigorous bounds [3]. We apply it to complete special-q boxes (Section 11).
- *Shape test.* Lenstra–Pomerance [33], used as a black box.

We make no novelty claim beyond saying that we did not find this combination, or a rigorous $L_N[1/2, c]$ with $c < 1$ (conditional or unconditional), in the literature.

1.6. **Organisation.** Section 2 fixes notation and lists the published results we use, with their exact ranges. Section 3 gives the parameters, the field law and the algorithm. Section 4 outlines the proof and gives a dependency diagram. The proof proper is in Sections 5–13, and Appendix A proves Lemma 8.4. Section 14 describes a toy implementation, and Section 15 lists the limitations.

2. Notation and published inputs

2.1. **Notation.** Throughout, log is the natural logarithm and

$$\mathcal{L} = \mathcal{L}_N := \sqrt{\log N \log\log N}, \qquad L = L_N[1/2, 1] = e^{\mathcal{L}}, \qquad L_N[1/2, c] = L^c.$$

We write $L^{o(1)}$ for any quantity of the form $\exp(o(\mathcal{L}))$, with the $o(1)$ uniform in N. Fixed powers L^{ε} and quantities $N^{o(1)}$ are *not* $L^{o(1)}$; when they occur, they are always absorbed by a fixed power of N. Constants $C, c, C_d, \ldots$ may depend on the fixed parameters (d, k, μ, η_0) and change from line to line. $(\log N)^{O(1)}$ denotes a fixed power of $\log N$.

$\omega(n)$ is the number of distinct prime factors of n, $\tau(n)$ the number of divisors, φ Euler's function and $P^+(n)$ the largest prime factor of n ($P^+(1) = 1$). $\pi(x)$ counts primes. For the smoothness bound B fixed in Section 3.1 we write

$$\begin{aligned} \Psi(x) &= \#\{1 \le n \le x : P^+(n) \le B\}, \qquad \varrho(x) = \Psi(x)/x, \\ \Psi(x; t, r) &= \#\{1 \le n \le x : P^+(n) \le B,\ n \equiv r\ (t)\}, \\ \Psi_t(x) &= \#\{1 \le n \le x : P^+(n) \le B,\ (n, t) = 1\}, \end{aligned}$$

and $\Psi(x; \chi) = \sum_{n \le x,\, P^+(n) \le B} \chi(n)$ for a Dirichlet character χ. When we need another smoothness bound y we write $\Psi(x, y)$, and so on. The set of nonzero B-smooth integers is $\mathcal{S}_B = \{n \in \mathbb{Z} \setminus \{0\} : P^+(|n|) \le B\}$. "Smooth" always means B-smooth.

$\boldsymbol{\mu}_2 = \boldsymbol{\mu}_2(N) = \{\epsilon \in (\mathbb{Z}/N)^\times : \epsilon^2 = 1\}$ and $Q_N = ((\mathbb{Z}/N)^\times)^2$. For odd N, each $(\mathbb{Z}/p^e)^\times$ is cyclic of even order, so $\boldsymbol{\mu}_2 \cong \{\pm 1\}^{\omega(N)}$ and $[(\mathbb{Z}/N)^\times : Q_N] = 2^{\omega(N)}$.

A *gcd check* of an integer x computes $g = \gcd(x, N)$. If $1 < g < N$, the current procedure stops and returns g as a proper factor (a *factor exit*). If $g = N$, the current object is rejected.

2.2. **Smooth numbers.**

Fact 2.1 (Canfield–Erdős–Pomerance [7])**.** *For each $\varepsilon > 0$, uniformly for $u \to \infty$ with $u \le (1 - \varepsilon) \log x / \log\log x$, $\Psi(x, x^{1/u}) = x\, u^{-u(1+o(1))}$.*

Fact 2.2 (Hildebrand–Tenenbaum [22, Thm. 3]; [29, Fact 3.15]). *Put $u = \log x/\log y$. There is $\alpha = \alpha(x,y)$ such that, uniformly for $x \ge y \ge 2$ and $1 \le c \le y$,*

$$\Psi(cx,y) = \Psi(x,y)\, c^{\alpha}\Big(1 + O\Big(\frac{1}{u} + \frac{\log y}{y}\Big)\Big), \qquad \alpha = \frac{\log(1+y/\log x)}{\log y}\Big(1 + O\Big(\frac{\log\log(y+1)}{\log y}\Big)\Big).$$

Fact 2.3 (Tenenbaum [51]; [29, Fact 3.16, Cor. 3.17]). *Fix $c > 0$. Uniformly for $2 \le y \le x$ and integers q with $\omega(q) \le y^{c/\log(1+u)}$,*

$$\Psi_q(x,y) = \frac{\varphi(q)}{q}\Psi(x,y)\Big(1 + O_c\Big(\frac{\log(1+u)\log(1+\omega(q))}{\log y}\Big)\Big)\Big(1 + O\Big(\frac{\omega(q)}{y}\Big)\Big).$$

We use these only in the following range. Fix $0 < \xi_1 < \xi_2$. We say x is *in range* if $N^{\xi_1} \le x \le N^{\xi_2}$, and we take $y = B = \lfloor L^a \rfloor$ with $a > 0$ fixed. Then $u = \log x/\log B \asymp (\log N/\log\log N)^{1/2}$, so $(\log x)^K < B < x^{1/\log\log x}$ for every fixed K and large N.

Lemma 2.4 (local behaviour of ϱ). *Let $C \ge 1$ be fixed. For $N \ge N_0(C, \xi_1, \xi_2, a)$ and $z_1 \le z_2 \le z_1 B^C$ with z_1 in range:*

(i) $\varrho(z_2) \le 2\varrho(z_1)$;
(ii) $\varrho(z_1) \le (\log N)^{O(C)}\varrho(z_2)$;
(iii) for fixed $Z \ge 1$, $\Psi(z_1(1+1/Z)) - \Psi(z_1) \ge \Psi(z_1)/(3Z)$, *and* $\Psi(z_1/2) \ge \Psi(z_1)/3$;
(iv) if $z = N^{\xi}L^{O(1)}$ with $\xi_1 \le \xi \le \xi_2$, then $\varrho(z) = L^{-\xi/(2a)+o(1)}$.

Proof. By Fact 2.2, $\log(1 + B/\log x) = \log B - \log\log x + O(\log x/B)$, so $(1-\alpha)\log B = \log\log x + O(\log\log B) = O(\log\log N)$; in particular $\alpha \to 1$. For $1 \le c \le B$, Fact 2.2 gives $\varrho(cx)/\varrho(x) = c^{\alpha-1}(1 + O(1/u))$, which lies between $B^{\alpha-1}(1 - O(1/u)) \ge (\log N)^{-O(1)}$ and $1 + O(1/u)$. Writing z_2/z_1 as a product of at most $C+1$ factors in $[1, B]$ gives (i) and (ii), since $u \to \infty$ uniformly in range. For (iii), $\Psi(z_1(1+1/Z)) - \Psi(z_1) = \Psi(z_1)\big((1+1/Z)^{\alpha}(1 + O(1/u)) - 1\big) \ge \Psi(z_1)(\alpha/(Z+1) - O(1/u))$. Also $\Psi(z_1/2) = 2^{-\alpha}\Psi(z_1)(1+O(1/u))$. For (iv), $u = (\xi\log N + O(\mathcal{L}))/(a\mathcal{L} + O(1)) = (\xi/a)(\log N/\log\log N)^{1/2} + O(1)$ and $\log u = \tfrac{1}{2}\log\log N(1+o(1))$, so Fact 2.1 gives $\log(1/\varrho(z)) = u\log u(1+o(1)) = (\xi/(2a))\mathcal{L}(1+o(1))$. □

2.3. **Other published inputs.**

Fact 2.5 (Lenstra–Pomerance [33]). *There is a Las Vegas algorithm which, for every integer $n > 1$, finds the complete prime factorisation of n in expected time at most $L_n[1/2, 1 + o(1)]$.*

Fact 2.6 (AKS [1]). *Primality of n can be decided deterministically in time $(\log n)^{O(1)}$.*

Fact 2.7 (large sieve; Montgomery–Vaughan [36, Thm. 1], see [35]). *For complex numbers a_F supported on K consecutive integers and $Q \ge 1$, $\sum_{m \le Q}\sum_{(b,m)=1,\, b \bmod m}\big|\sum_F a_F e(bF/m)\big|^2 \le (K + Q^2)\sum_F |a_F|^2$.*

Fact 2.8 (Brun–Titchmarsh; Montgomery–Vaughan [36, Thm. 2]). *For $1 \le q < y$ and $(a,q) = 1$, $\pi(x+y; q, a) - \pi(x; q, a) < 2y/(\varphi(q)\log(y/q))$.*

The inputs for the restricted Bombieri–Vinogradov estimate (Drappeau's Lemme 5, de la Bretèche–Tenenbaum's local bound, and the multiplicative large sieve) are stated in Appendix A. The number-field inputs (Lagarias–Odlyzko, Stark, Landau) are stated in Section 10.

3. Parameters, field law and the algorithm

3.1. **Parameters.** *Fixed parameters.* An odd integer $d \geq 3$; an integer $k > (d+1)/4$; $\nu := 1/(4k)$; a rational μ with $\nu + \eta_0 \leq \mu < 1$; a rational $\eta_0 > 0$. Put

$$h := h_*(\mu) + \eta_0, \qquad h_*(\mu) = \max\big(\underbrace{2\nu}_{\text{kernel}}, \underbrace{\mu - (d-2)\nu}_{\text{level } 1/2}, \underbrace{1 - d\mu + \nu}_{\text{window}}\big), \qquad \theta := 2\mu + h + (d-1)\nu,$$

and assume $\theta < 1$. Put $a := \theta/\sqrt{2}$.

Quantities depending on N (the integer handed to the splitting routine, see Section 13):

- $B := \lfloor L^a \rfloor$, the smoothness bound;
- $X := \lfloor \sqrt{N/2} \rfloor$; j maximal with $2^{kj} \leq X$; $Y := 2^{j-1}$; $J := \{\ell \text{ prime} : Y < \ell \leq 2Y\}$; $\pi_J := \#J$;
- $S := \lceil L^{1/(2\sqrt{2})} \rceil$, so that $S^2 = L^{\theta/(2a)+o(1)}$;
- for $\ell \in J$: $V_\ell := \lfloor \sqrt{\ell}\, S \rfloor$, with envelopes $V_- := \lfloor \sqrt{Y}\, S \rfloor \leq V_\ell \leq V_{\max} := \lfloor \sqrt{2Y}\, S \rfloor$; and $T_\ell := dHV_\ell^{d-1}$;
- $M := \lfloor N^\mu \rfloor$, $H := \lfloor N^h \rfloor$, $W := \lceil N^{\mu+\nu+\eta_0/2} \rceil$;
- $X^* := C'_d M H V_{\max}^d$ and $X_r := 2MV_{\max}/Y$, where $C'_d := 2^{2d+5} + 2d$; and $\rho_r := \varrho(X_r)$, $\rho_F := \varrho(X^*)$.

Since $2^{k(j+1)} > X$, we have $X^{1/k}/4 < Y \leq X^{1/k}/2$, so $Y = N^{2\nu+o(1)}$ and every product of k primes of J is at most X. $V_\ell = N^\nu L^{1/(2\sqrt{2})+o(1)}$, $X_r = N^{\mu-\nu} L^{O(1)}$ and $X^* = N^{\mu+h+d\nu} L^{O(1)}$. By Lemma 2.4(iv),

(2)
$$\rho_r \rho_F = L^{-(\mu-\nu+\mu+h+d\nu)/(2a)+o(1)} = L^{-\theta/(2a)+o(1)} = L^{-1/\sqrt{2}+o(1)}, \qquad \text{so} \qquad S^2 \rho_r \rho_F = L^{o(1)}.$$

Lemma 3.1 (size relations)**.** *For $N \geq N_0(d, k, \mu, \eta_0)$:*

(a) $B < 2S^2 < Y < \ell$ for $\ell \in J$; $S < \sqrt{Y}$; $V_\ell < \ell$; and $B < V_\ell/2$.
(b) $2MV_{\max} < W$, $W/(2MV_{\max}) \geq N^{\eta_0/3}$, and $2W < N$.
(c) $M/V_{\max} \geq N^{\eta_0/2}$ and $H/V_{\max}^2 \geq N^{\eta_0/2}$.
(d) $M(H+1) < N$ and $2H + 1 < N$.
(e) $2X^2 < N$.

Proof. (a) $B = L^{\theta/\sqrt{2}} \leq L^{1/\sqrt{2}} < 2S^2$ since $\theta < 1$; $S^2 = L^{O(1)}$ while $Y = N^{2\nu+o(1)}$; $V_\ell \leq \sqrt{\ell} S < \ell$; $V_\ell/2 \geq N^{\nu+o(1)} > B$. (b) $2MV_{\max} = N^{\mu+\nu} L^{O(1)}$. Also $\mu + \nu + \eta_0/2 \leq \theta < 1$, because $\mu \geq \nu$, $h \geq 2\nu + \eta_0$ and $d \geq 3$. (c) $\mu - \nu \geq \eta_0$ and $h - 2\nu \geq \eta_0$. (d) $\mu + h = \theta - \mu - (d-1)\nu < \theta < 1$. (e) N is odd in our application, so $2X^2 \leq N - 1$. □

At $d = 9$, $k = 13$ the exponents are $\nu = 1/52$, $\mu = 17/156$, $h = 1/26 + \eta_0$; the rational cofactor has exponent $\mu - \nu = 14/156$ and the algebraic value $\mu + h + d\nu = 59/156 + \eta_0$.

Lemma 3.2 (optimal μ)**.** *At $\eta_0 \downarrow 0$, the minimum of θ over $\mu \in [\nu, 1)$ is attained at $\mu = (1-\nu)/d$ with $\theta = 2/d + (d^2 + d - 2)\nu/d$ if $k < (d^2+1)/4$, and at $\mu = (1 + (d-1)\nu)/(d+1)$ with $\theta = 3/(d+1) + (4d-2)\nu/(d+1)$ if $k \geq (d^2+1)/4$.*

Proof. At zero slack $\theta(\mu) = \max\big(2\mu + (d+1)\nu,\ 3\mu + \nu,\ 1 + (2-d)\mu + d\nu\big)$. The first two are increasing and the third decreasing, so the minimum is at the intersection of the third with the larger of the first two. The third meets the first at $\mu = (1-\nu)/d$ and the second at $\mu = (1 + (d-1)\nu)/(d+1)$. Comparing the first two there, the first is the larger iff $\mu \leq d\nu$, which at $\mu = (1-\nu)/d$ means $k < (d^2+1)/4$. □

3.2. **The field law.**

Lemma 3.3 (generalised base expansion; cf. [6, §3])**.** *Let $g_1 \geq 2$, $\gcd(g_0, g_1) = 1$, $\gcd(g_1, N) = 1$, and let w be a prime with $w > g_1$. Put $T_d := Nw$ and, for $i = d, d-1, \dots, 1$: let a_i be the integer with $|a_i| \leq g_1/2$ and $a_i \equiv T_i g_0^{-i} \pmod{g_1}$, and $T_{i-1} := (T_i - a_i g_0^i)/g_1$. Put $a_0 := T_0$. Then the form $\hat{F}(x,y) := \sum_{i=0}^{d} a_i x^i y^{d-i}$ satisfies $\hat{F}(g_0, g_1) = Nw$. If $g_0, g_1 \in [M/2, M]$, then $|a_0| \leq 2^d Nw/M^d + d2^{d-1}M$, and $a_d \not\equiv 0 \pmod{g_1}$.*

Proof. $T_i \equiv a_i g_0^i \pmod{g_1}$, so $T_{i-1} \in \mathbb{Z}$, and by induction $T_i = \sum_{j \leq i} a_j g_0^j g_1^{i-j}$. At $i = d$ this is $\hat{F}(g_0, g_1) = Nw$. Next, $a_0 = (Nw - \sum_{i \geq 1} a_i g_0^i g_1^{d-i})/g_1^d$, with $Nw/g_1^d \leq 2^d Nw/M^d$ and $|a_i|(g_0/g_1)^i \leq (M/2)2^i$. Finally $a_d g_0^d \equiv Nw \pmod{g_1}$, and Nw is a unit modulo g_1 because $\gcd(g_1, N) = 1$ and w is a prime larger than g_1. □

Definition 3.4 (field law $\mathcal{A}$)**.** Let $\mathcal{G} := \{(g_0, g_1) \in \mathbb{Z}^2 \cap [M/2, M]^2 : \gcd(g_0, g_1) = 1\}$ and $C_H := [-H, H]^d \cap \mathbb{Z}^d$. A *field* is $A = (g, w, c)$ with three independent components: $g = (g_0, g_1)$ uniform on $\mathcal{G}$; w uniform among the primes in $(W, 2W]$; c uniform on C_H. Put

$$G(x,y) = G_g(x,y) := g_1 x - g_0 y, \quad R_c(x,y) := \sum_{i=0}^{d-1} c_i x^{d-1-i} y^i, \quad F_A := \hat{F}_{g,w} + G \cdot R_c,$$

where $\hat{F}_{g,w}$ is Lemma 3.3 applied to Nw, and $f = f_A(x) := F_A(x, 1) = \sum_i f_i x^i$. For $P = (a, b)$ put $v(P) := (a^{d-1-i} b^i)_{0 \leq i < d}$, so that $F_A(P) = \hat{F}(P) + G(P)\langle c, v(P)\rangle$.

So $F_A(g_0, g_1) = Nw$, $f_d = a_d + g_1 c_0 \equiv a_d \not\equiv 0 \pmod{g_1}$, and if $\gcd(g_1, N) = 1$ the map $x \mapsto m := g_0 g_1^{-1} \bmod N$ kills f, because $f(m) \equiv g_1^{-d} F_A(g_0, g_1) = g_1^{-d} Nw \equiv 0$. Thus $(f, g_1 x - g_0)$ is a number field sieve pair for N, with resultant $\pm Nw$.

Lemma 3.5 (sizes)**.** *Let $N \geq N_0$, A a field, $\ell \in J$, $P = (a, b)$ with $|a|, |b| \leq V_\ell$, and $|z| \leq T_\ell$. Then $|G(P)| \leq 2MV_\ell$, $|\hat{F}(P)| \leq C_d M H V_\ell^d N^{-\eta_0/2}$ with $C_d = 2^{2d+5}$, and $|\hat{F}(P) + G(P)z| \leq X^*$. In particular $|F_A(P)| \leq X^*$.*

Proof. $|G(P)| \leq |g_1 a| + |g_0 b| \leq 2MV_\ell$. By Lemma 3.3, $|\hat{F}(P)| \leq (dM/2 + |a_0|)V_\ell^d$ and $|a_0| \leq 2^d Nw/M^d + d2^{d-1}M$. Since $w \leq 2W \leq 3N^{\mu+\nu+\eta_0/2}$ and $M \geq N^\mu/2$, $Nw/M^d \leq 3 \cdot 2^d N^{1-(d-1)\mu+\nu+\eta_0/2}$, which is at most $12 \cdot 2^d MHN^{-\eta_0/2}$ because $h \geq 1 - d\mu + \nu + \eta_0$. Also $d2^d M \leq MHN^{-\eta_0/2}$ for large N. This gives the bound on $\hat{F}$. Finally $|G(P)z| \leq 2MV_\ell \cdot dHV_\ell^{d-1}$, and $|\langle c, v(P)\rangle| \leq dHV_\ell^{d-1} = T_\ell$. □

Lemma 3.6 (homogeneity)**.** *For every field and every P, $g_1^d F_A(P) \equiv Nw\, b^d \pmod{G(P)}$, and $F_A(P) \equiv \hat{F}(P) \pmod{G(P)}$.*

Proof. $g_1 P - b(g_0, g_1) = (G(P), 0)$, and $F_A(x_1 + G, x_2) \equiv F_A(x_1, x_2) \pmod{G}$ because F_A has integer coefficients. Hence $g_1^d F_A(P) = F_A(g_1 P) \equiv F_A(bg_0, bg_1) = b^d F_A(g_0, g_1) = b^d Nw$. The second claim holds because $F_A - \hat{F} = G \cdot R_c$. □

3.3. **Candidates and the counted relations.** For $\ell \in J$ and a field A:

- $\mathcal{P}_\ell := \{P = (a, b) \in \mathbb{Z}^2 :\ b \text{ prime},\ V_\ell/2 < b \leq V_\ell,\ |a| \leq V_\ell,\ \gcd(a, b) = 1\}$;
- $\Lambda_\ell(g) := \{P \in \mathbb{Z}^2 : \ell \mid G_g(P)\}$, a lattice of index ℓ (since $\gcd(g_0, g_1) = 1$, $g \not\equiv 0 \bmod \ell$);
- the *filtered count*

  $$N_\ell(A) := \#\Big\{P \in \mathcal{P}_\ell \cap \Lambda_\ell(g) :\ \tfrac{1}{32} MV_\ell \leq |G(P)|,\ \ G(P)/\ell \in \mathcal{S}_B,\ \ F_A(P) \in \mathcal{S}_B\Big\},$$

  and $N_\ell(A) := 0$ if $\ell \mid N$;

- the set of *descendable primes* $\mathcal{D}_A := \{\ell \in J : N_\ell(A) \geq 1\}$, $\delta_A := |\mathcal{D}_A|/\pi_J$, and $\gamma := \mathbb{E}_{A\sim\mathcal{A}}\delta_A = \Pr_{A,\ell}[N_\ell(A) \geq 1]$, with ℓ uniform on J.

The upper filter $|G(P)| \leq 2MV_\ell$ holds automatically (Lemma 3.5). For a counted P we call (P, ℓ) an *atom*. Writing $G(P) = \sigma_P \ell n_P$ with $\sigma_P = \pm 1$ and $n_P > 0$ smooth, an atom certifies $\ell \equiv \sigma_P G(P)/n_P \pmod N$ together with the smoothness of $F_A(P)$.

Lemma 3.7 (regularity; cf. [40])**.** *Let λ_1 be the length of a shortest nonzero vector of $\Lambda_\ell(g)$. If $\lambda_1 < \sqrt{\ell}/(2S)$, then $\Lambda_\ell(g) \cap \mathcal{P}_\ell = \emptyset$. Otherwise $\#(\Lambda_\ell(g) \cap [-V_\ell, V_\ell]^2) \leq 200S^2$.*

Proof. Write $V = V_\ell$. Let w_0 be a shortest vector, with content $e \leq |w_0| < \ell$. Then e is invertible modulo ℓ, so $w_0/e \in \Lambda_\ell(g)$, and minimality gives $e = 1$. Thus w_0 is primitive in $\Lambda_\ell(g)$, and the lattice is the union of the lines $jw' + \mathbb{Z}w_0$ ($j \in \mathbb{Z}$), parallel to w_0 at spacing $\ell/|w_0| > 2S\sqrt{\ell} \geq 2V$. Every point of $[-V, V]^2$ has norm at most $\sqrt{2}V < 2V$, so it lies on $\mathbb{Z}w_0$. The only primitive points there are $\pm w_0$, and $|w_0| < \sqrt{\ell}/(2S) < V/2 < b$ for every $(a, b) \in \mathcal{P}_\ell$. In the regular case take a reduced basis $\lambda_1 \leq \lambda_2$ with $\lambda_1\lambda_2 \leq (2/\sqrt{3})\ell$. Translates of its fundamental parallelogram at the lattice points of $[-V, V]^2$ are disjoint and lie in a square of side $2V + 4\lambda_2$. Since $V^2 \leq \ell S^2$ and $\lambda_2^2 \leq (4/3)\ell^2/\lambda_1^2 \leq (16/3)S^2\ell$, the count is at most $(2V+4\lambda_2)^2/\ell \leq (8V^2+32\lambda_2^2)/\ell \leq 200S^2$. □

Lemma 3.8 (coprimality and the forced residue)**.** *Let $N \geq N_0$ have no prime factor $\leq B$, let $\ell \in J$ with $\ell \nmid N$, and let $P = (a, b) \in \mathcal{P}_\ell \cap \Lambda_\ell(g)$ with $n := G(P)/\ell \in \mathcal{S}_B$. Put $t := |G(P)|$. Then $\gcd(t, bg_1Nw) = 1$ and*

$$\hat{F}_{g,w}(P) \equiv c_P\, w \pmod t, \qquad c_P := Nb^d g_1^{-d} \in (\mathbb{Z}/t)^\times, \tag{3}$$

where c_P depends on (ℓ, P, g) but not on w.

Proof. b is a prime with $B < V_\ell/2 < b \leq V_\ell < \ell$ (Lemma 3.1(a)), so $b \nmid \ell|n| = t$. A prime dividing g_1 and $t = |g_1a - g_0b|$ divides g_0b, hence b (as $\gcd(g_0, g_1) = 1$), which is impossible. N has no prime factor $\leq B$, n is smooth and $\ell \nmid N$, so $\gcd(t, N) = 1$. Finally $w > W > 2MV_\ell \geq t$ is prime. Now apply Lemma 3.6. □

Remark 3.9. Without the hypothesis that $G(P)/\ell$ is smooth, (3) can fail (for example $\gcd(g_1, t) = b$ is possible). Such candidates never enter the moment sums.

3.4. **The algorithm.**

Algorithm 3.10 (FACTOR(N))**.**

1. If $N = 1$ return the empty list. If N is prime (AKS), return $[N]$. If $N = m^e$ with $e \geq 2$ (integer roots, $e \leq \log_2 N$), return FACTOR(m) with all exponents multiplied by e.
2. Trial-divide N by all primes $\leq B(N)$. Record the primes found, and let N' be the cofactor. If $N' = 1$, stop. If N' is prime or a perfect power, handle it as in step 1.
3. Run SPLIT(N') and trial division of N' in parallel, alternating steps, until one returns a proper factor g of N'. (This alternation removes the need for any branch "$N' < N_0$": the algorithm then uses no analytic constant at all, and pays at most a factor 2 in expected time; see Remark A.17.)
4. Return FACTOR(g) merged with FACTOR(N'/g), together with the primes from step 2.

Algorithm 3.11 (SPLIT(N); N odd, all prime factors $> B$, not a prime power)**.**

1. *Field selection.* Repeat: $A \leftarrow$ SAMPLE_FIELD; if A is rejected, continue; row $\leftarrow$ DESCEND_ONCE(A); if a row is returned, keep A and go to step 2. (Any factor exit returns the factor.)
2. *Rows.* Call DESCEND_ONCE(A) until M_{rows} rows are held, where M_{rows} is defined in (4).

3. *Post-processing.* Run POSTPROCESS(A, rows) (Algorithm 10.14). It returns a nonempty set S of rows and $\mathsf{Y_S} \in (\mathbb{Z}/N)^\times$. Compute $\mathsf{X_S} := \prod_{i\in\mathsf{S}} r_i \bmod N$ and $g := \gcd(\mathsf{X_S} - \mathsf{Y_S}, N)$. If $1 < g < N$ return g.
4. Otherwise discard all rows and go to step 2 (same field, fresh hidden roots).

Algorithm 3.12 (SAMPLE_FIELD)**.** 1. Draw (g_0, g_1) uniformly from $[M/2, M]^2 \cap \mathbb{Z}^2$ until $\gcd(g_0, g_1) = 1$. Draw a uniform prime $w \in (W, 2W]$ (uniform integers and AKS). Draw c uniform in C_H. Gcd-check g_1, w and then $f_d = a_d + g_1 c_0$. (All three are nonzero and smaller than N, by Lemma 3.1, so these checks either exit with a factor or pass.)
2. Compute f. Let e be its content. Then $e \mid F_A(g_0, g_1) = Nw$ and $e \le |f_d| < N$. Gcd-check e. If it passes, $e \in \{1, w\}$. If $e = w$, *reject*.
3. *Irreducibility certificate* (Lemma 10.1): if no prime $r \in (r_0(d), \frac{1}{2}\log H]$ has $f \bmod r$ of degree d and irreducible over $\mathbb{F}_r$, *reject*.
4. Gcd-check $f_d^{d-2} f'(m)$, where $m = g_0 g_1^{-1} \bmod N$; if $N \mid f'(m)$, *reject*.
5. Return A.

Algorithm 3.13 (DESCEND_ONCE(A))**.** 1. Draw r uniform in $[1, N-1]$; gcd-check r (a non-unit r gives a factor exit). Put $R := r^2 \bmod N$.
2. Let (u_0, v_0) be a shortest nonzero vector of $\Lambda_R := \{(u, v) \in \mathbb{Z}^2 : u \equiv Rv \pmod N\}$ (Gauss reduction, ties broken by a fixed rule). Gcd-check u_0 and v_0.
3. *Shape test.* Factor $|u_0|$ and $|v_0|$ completely with the algorithm of Fact 2.5, and certify each output prime with AKS. Require that $|u_0|$ and $|v_0|$ are each a product of at most k *distinct* primes, all in J; otherwise return FAIL.
4. For each prime $\ell \mid u_0 v_0$, gcd-check ℓ and run SQ_SEARCH(A, ℓ) (cached per (A, ℓ)). If any search fails, return FAIL.
5. Return the row $(r, R, \operatorname{sgn}(u_0 v_0), (e_\ell)_\ell, (P_\ell)_\ell)$, where $e_\ell = +1$ if $\ell \mid u_0$, $e_\ell = -1$ if $\ell \mid v_0$, and P_ℓ is the atom found for ℓ, with the factorisations of n_{P_ℓ} and $F_A(P_\ell)$.

Algorithm 3.14 (SQ_SEARCH(A, ℓ); a deterministic function of (A, ℓ))**.** 1. Gauss-reduce a basis of $\Lambda_\ell(g)$. If $\lambda_1 < \sqrt{\ell}/(2S)$, return FAIL (Lemma 3.7).
2. Enumerate $\Lambda_\ell(g) \cap [-V_\ell, V_\ell]^2$ (at most $200S^2$ points). Keep the points $P \in \mathcal{P}_\ell$ (b prime by AKS) with $G(P) \neq 0$, $F_A(P) \neq 0$ and $|G(P)| \ge MV_\ell/32$. Gcd-check $G(P)$ and $F_A(P)$ for each.
3. Compute the B-smooth parts of all the numbers $|G(P)/\ell|$ and $|F_A(P)|$ with one batch (Proposition 11.2).
4. Return the first kept P, in a fixed order, for which both numbers are B-smooth, together with the factorisations of its two values (trial division by the primes $\le B$). If there is none, return FAIL.

POSTPROCESS is Algorithm 10.14 in Section 10. The number of rows is

$$M_{\text{rows}} := C_{\text{col}} + 12, \qquad C_{\text{col}} := 2 + \pi(B) + \#\{(r, s) : r \le B \text{ prime},\ r \nmid D,\ f(s) \equiv 0\ (r)\} + t_\chi, \tag{4}$$

where $D = f_d \operatorname{disc}(f)$ and $t_\chi = O_d(\log N)$ is defined in Lemma 10.11. So $M_{\text{rows}} \le (d+1)\pi(B) + O_d(\log N)$.

Remark 3.15 (on the design)**.** Each design choice serves one step of the proof.

- *b prime in $(V/2, V]$.* This makes every moving residue reduced (Lemma 3.8), removes small b, and gives the effective wedge bound in Lemma 9.2.
- *The filter $|G(P)| \ge MV/32$.* Candidates with small $|G|$ have algebraic values smaller by a power of N. Their smoothness probability could then exceed ρ_F by $L^{\Theta(1)}$, which would spoil the second moment. The first moment only uses $|G| \asymp MV$, so the filter costs nothing.

- *Two-dimensional rational side.* With (g_0, g_1) random, the rational values of a fixed candidate fill whole fibres of a linear map. The rational part of the first moment is then an interval count, and the partner's rational weight in the second moment can be bounded by divisor switching along a fibre (Step 5 of Theorem 9.7).
- *The multiplier w.* By Lemma 3.6 every algebraic value of a candidate lies in one class modulo $G(P)$, and that class is determined by the rational data. With $F(g_0, g_1) = N$ this class moves together with the partner's rational indicator, and we know no way to decouple them. Proposition 3.16 shows that no randomisation of the coefficients at a fixed resultant can move it. Multiplying the resultant by an independent random prime $w > t$ makes the class $c_P w$ almost uniform (Lemma 9.3).
- *The windows.* The window floor $h \geq 1 - d\mu + \nu + \eta_0$ keeps the large constant coefficient $a_0 \approx Nw/M^d$ invisible at the scale MHV^d (Lemma 3.5). The level floor keeps the moduli $|G(P)| \approx N^{\mu+\nu}$ below the square root of the algebraic values, as Lemma 8.4 requires. The kernel floor $H \geq V^2$ is needed by Lemma 9.2.

Proposition 3.16 (no coefficient-only repair; cf. [29])**.** *If two integral binary forms of degree d take the same value at the primitive point (g_0, g_1), their difference is divisible by $G = g_1 x - g_0 y$ in $\mathbb{Z}[x, y]$. Consequently every form F of degree d with $F(g_0, g_1) = N$ satisfies $g_1^d F(P) \equiv N b^d \pmod{G(P)}$ for all P.*

Proof. The difference D vanishes at (g_0, g_1), so G divides it over $\mathbb{Q}$; G is primitive, so by Gauss's lemma it divides it over $\mathbb{Z}$. The congruence is Lemma 3.6 with $w = 1$. □

4. Overview of the proof

The proof of Theorem 1.1 has three parts. Figure 1 shows how they depend on each other.

Part I: from relation supply to a factor. Suppose the following averaged statement holds.

$$\gamma = \Pr_{A \sim \mathcal{A},\ \ell \in J}\big[N_\ell(A) \geq 1\big] \ \geq \ L^{-o(1)}. \tag{D_k}$$

- Reverse Markov gives many fields with $\delta_A \geq \gamma/4$.
- In such a field the counting lemma (Theorem 5.2) shows that a uniformly random square R splits as u/v with u, v products of k descendable primes with probability $L^{-o(1)}$ (Theorem 7.2).
- The field-selection rule keeps a field with law proportional to its success probability. A harmonic-mean identity, combined with the factor-exit bookkeeping, bounds the expected cost of all the rows (Proposition 7.3).
- The rows are combined by the post-processing of Section 10. Its output depends only on the squares R_i and on independent coins, so Theorem 6.1 makes each round fruitful with probability $\geq 1/2$.

Part II: relation supply. We prove (D_k) by the Paley–Zygmund inequality (Theorem 9.9). Two moment bounds are needed, and averaging over ℓ is essential in both.

- The first moment (Theorem 8.7): $\mathbb{E}_{\ell,A} N_\ell(A) \geq S^2 \rho_r \rho_F / (\log N)^{O(1)}$. It uses the one-point smoothing Lemma 8.1 and the restricted Bombieri–Vinogradov estimate Lemma 8.4.
- The second moment (Theorem 9.7): $\mathbb{E}_{\ell,A}[N_\ell(N_\ell - 1)] \leq (\log N)^{O(1)} S^4 \rho_r^2 \rho_F^2$. It uses Lemma 9.2 (kernel), Lemma 9.3 (multiplier), Lemma 9.4 (divisor switching) and Theorem 9.5 (large sieve).

Part III: the algorithmic side. This covers exact recognition of $N_\ell(A)$ by batch smoothness (Section 11), rejection of bad fields (Lemma 10.1), rigorous post-processing at fixed degree (Section 10), the cost assembly (Section 12), and uniformity in N (Section 13).

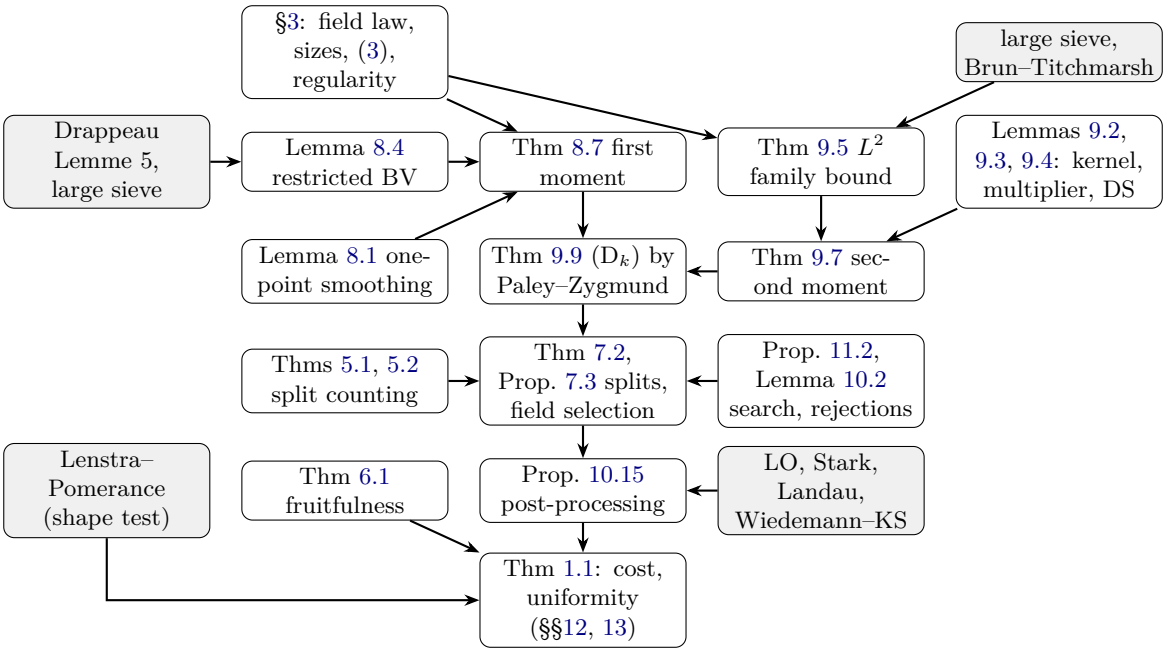


FIGURE 1. Dependencies. Shaded boxes are published results, used only through their statements. Drappeau's Lemme 5 is published, but its proof refers to computations in Harper's unpublished preprint (Appendix A).

Remark 4.1 (two interfaces that need care)**.** (a) *Order of operations.* The moment bounds are proved under the raw product law $\mathcal{A}$. Rejection of bad fields is then accounted for at the level of the bounded quantity $\delta_A \in [0,1]$, as $\mathbb{E}[\delta_A 1_{\mathcal{R}^c}] \geq \gamma - \Pr(\mathcal{R})$, and only after that is the field-selection rule applied. Conditioning on acceptance before the moment estimates would destroy the product structure that the second moment uses.

(b) *Computable S.* The algorithm cannot compute Ψ, so S is an explicit function of N and $\mathbb{E}N_\ell$ need not be ≥ 1. The Paley–Zygmund form $\gamma \geq (\mathbb{E}N)^2/(\mathbb{E}N(N-1)+\mathbb{E}N)$ needs neither $\mathbb{E}N \geq 1$ nor a polylogarithmic lower bound on γ. Indeed $S^2\rho_r\rho_F$ is $L^{o(1)}$ but, by de Bruijn's second-order term, smaller than every negative power of $\log N$. So only $\gamma \geq L^{-o(1)}$ holds, and that suffices because k is fixed.

5. Split counting

In this section N is odd and $h_2 := 2^{\omega(N)} = [(\mathbb{Z}/N)^\times : Q_N]$. For a set $\mathcal{S}$ of integers coprime to N, let $E_{\mathcal{S}}$ be the event that $u \equiv Rv \pmod N$ for some $u, v \in \mathcal{S}$, where R is uniform on Q_N.

Theorem 5.1 (split counting; ratio form of the Dixon–Pomerance lemma)**.** *Let $Y' \geq 2$ with $2Y'^2 < N$, and let $\mathcal{S} \subset [Y'/2, Y'] \cap \mathbb{Z}$ consist of integers coprime to N. Then*

$$\Pr[E_{\mathcal{S}}] \;\geq\; \frac{|\mathcal{S}|^4}{2h_2\varphi(N)Y'^2(1+\log Y')}.$$

Moreover, if $(u,v) \in \mathcal{S}^2$ satisfies $u \equiv Rv$, then (u,v) is an integer multiple of a shortest nonzero vector (u_0, v_0) of $\Lambda_R = \{(x,y) \in \mathbb{Z}^2 : x \equiv Ry \pmod N\}$.

Proof. Put $n(R) := \#\{(u,v) \in \mathcal{S}^2 : u \equiv Rv\}$ for $R \in Q_N$.

First moment. Let $\widehat{G}$ be the group of characters of $(\mathbb{Z}/N)^\times$ trivial on Q_N, so $|\widehat{G}| = h_2$, every $\chi \in \widehat{G}$ is real, and $1[x \in Q_N] = h_2^{-1} \sum_{\chi \in \widehat{G}} \chi(x)$. Then

$$\sum_{R \in Q_N} n(R) = \#\{(u,v) \in \mathcal{S}^2 : uv^{-1} \in Q_N\} = h_2^{-1} \sum_{\chi \in \widehat{G}} \Big(\sum_{u \in \mathcal{S}} \chi(u) \Big)^2 \;\geq\; |\mathcal{S}|^2 / h_2,$$

by keeping only the trivial character.

Second moment. If $u \equiv Rv$ and $u' \equiv Rv'$ then $uv' \equiv u'v \pmod N$ and $|uv' - u'v| < Y'^2 < N$, so $uv' = u'v$: the two vectors are proportional. Write $g = \gcd(u,v)$ and $(u,v) = g(u_1, v_1)$. The vectors of $\mathcal{S}^2$ proportional to (u,v) are of the form $j(u_1, v_1)$ with $ju_1 \in [Y'/2, Y']$, and there are at most $Y'/(2u_1) + 1 \leq 2g$ of them. Hence

$$\sum_R n(R)^2 \leq \sum_{(u,v) \in \mathcal{S}^2} 2 \gcd(u,v) \leq 2 \sum_{g \geq 1} g \lfloor Y'/g \rfloor^2 \leq 2Y'^2 (1 + \log Y').$$

Cauchy–Schwarz. $\#\{R : n(R) > 0\} \geq (\sum n)^2 / \sum n^2$. Dividing by $|Q_N| = \varphi(N)/h_2$ gives the bound.

Shortest vector. Λ_R has determinant N. Let $(u,v) \in \mathcal{S}^2 \cap \Lambda_R$ and let (u_0, v_0) be a shortest vector. Both have Euclidean norm at most $\sqrt{2}Y'$, so $|uv_0 - u_0 v| \leq 2Y'^2 < N$ by Cauchy–Schwarz. The determinant is a multiple of N, hence 0, and the vectors are proportional. A shortest vector is primitive in Λ_R, so $(u,v) \in \mathbb{Z}(u_0, v_0)$. □

Theorem 5.2 (k-products)**.** *Let $\mathcal{D}$ be a set of primes in a dyadic interval, all coprime to N, with $n := |\mathcal{D}| \geq 4k^2$. Let $\mathcal{S}$ be a set of products of exactly k distinct primes of $\mathcal{D}$, contained in a window $[Y'/2, Y']$ with $2Y'^2 < N$, and with $|\mathcal{S}| \geq \binom{n}{k}/(2k)$. Then*

$$\Pr[E_{\mathcal{S}}] \;\geq\; \frac{|\mathcal{S}|^2}{c_k h_2 \varphi(N)}, \qquad c_k := 8k^2 \binom{2k}{k}.$$

Proof. As before $\sum_R n(R) \geq |\mathcal{S}|^2/h_2$ and $\sum_R n(R)^2 \leq \sum_{(u,v) \in \mathcal{S}^2} \#\{\text{partners of } (u,v)\}$, where the partners are the elements of $\mathcal{S}^2$ proportional to (u,v). Classify (u,v) by the number i of primes that u and v share. Then $(u,v) = (ju_1, jv_1)$ with j the product of the i shared primes. There are at most $\binom{n}{i}\binom{n}{k-i}^2$ such pairs, and each has at most $\binom{n}{i}$ partners $j'(u_1, v_1)$. Now

$$\binom{n}{i}\binom{n}{k-i} = \binom{k}{i}\binom{n}{k} \prod_{t<i} \frac{n-t}{n-k+i-t} \leq \binom{k}{i}\binom{n}{k}\Big(1 + \frac{k}{n-k}\Big)^k \leq \sqrt{2}\binom{k}{i}\binom{n}{k},$$

since $k^2/(n-k) \leq 1/3$ for $n \geq 4k^2$ and $e^{1/3} < \sqrt{2}$. So the i-th class contributes at most $2\binom{k}{i}^2\binom{n}{k}^2 \leq 8k^2\binom{k}{i}^2 |\mathcal{S}|^2$. Summing, with $\sum_i \binom{k}{i}^2 = \binom{2k}{k}$, gives $\sum n(R)^2 \leq c_k |\mathcal{S}|^2$, and Cauchy–Schwarz finishes the proof as before. □

Remark 5.3. No independence, no distribution of the primes and no character sum is used. The square-class condition is absorbed by positivity, which is Dixon's coset pigeonhole written with characters. The factor $2^{\omega(N)}$ is harmless after trial division (Lemma 5.4).

Lemma 5.4 (small factors)**.** *If N has no prime factor $\leq B$, then $\omega(N) \leq \log N / \log B$ and $2^{\omega(N)} = L^{o(1)}$.*

Proof. $\omega(N) \log B \leq \log N$, and $\log N / \log B \asymp (\log N / \log\log N)^{1/2} = o(\mathcal{L})$. □

6. Fruitfulness: Dixon descent

Theorem 6.1 (Dixon descent; Rabin/Dixon fibre symmetry)**.** *Let N be odd with $\omega(N) \geq 2$. Let $r_1, r_2, \ldots$ be independent and uniform on $[1, N-1]$, let $R_i := r_i^2 \bmod N$, and let ξ be further randomness independent of (r_i). Let $\mathcal{F} := \sigma((R_i)_i, \xi)$. Let $\mathcal{E} \in \mathcal{F}$ be an event on which an $\mathcal{F}$-measurable nonempty finite set S of indices with every R_i ($i \in \mathsf{S}$) a unit, and an $\mathcal{F}$-measurable $\mathsf{Y} \in (\mathbb{Z}/N)^\times$, are given, with $\mathsf{X}^2 \equiv \mathsf{Y}^2 \pmod N$ where $\mathsf{X} := \prod_{i \in \mathsf{S}} r_i$. Then, conditional on $\mathcal{F}$, on $\mathcal{E}$ the ratio X/Y is uniform on $\boldsymbol{\mu}_2$, and*

$$\Pr\big[1 < \gcd(\mathsf{X} - \mathsf{Y}, N) < N \,\big|\, \mathcal{F}\big] = 1 - 2^{1-\omega(N)} \geq \tfrac{1}{2} \qquad \text{on } \mathcal{E}.$$

Proof. For every unit square R fix a root $\rho(R)$. Conditional on R_i (a unit), $r_i = \epsilon_i \rho(R_i)$ with ϵ_i uniform on $\boldsymbol{\mu}_2$, because each unit square has exactly $|\boldsymbol{\mu}_2|$ roots and r_i is uniform. These ϵ_i are independent of each other and of ξ. So, conditional on $\mathcal{F}$, they are i.i.d. uniform on $\boldsymbol{\mu}_2$. On $\mathcal{E}$, $\mathsf{X}/\mathsf{Y} = \prod_{i \in \mathsf{S}} \epsilon_i \cdot \prod_{i \in \mathsf{S}} \rho(R_i) \mathsf{Y}^{-1}$, where the second factor is $\mathcal{F}$-measurable and lies in $\boldsymbol{\mu}_2$ (both X/Y and the first factor do). A product of independent uniform elements of a finite group over a nonempty index set is uniform, so X/Y is uniform on $\boldsymbol{\mu}_2$. Finally, for $\zeta \in \boldsymbol{\mu}_2$ and the factorisation $N = \prod p_j^{e_j}$ we have $\zeta \equiv \pm 1 \pmod{p_j^{e_j}}$ for each j, and $\zeta - 1$ is a unit modulo $p_j^{e_j}$ when $\zeta \equiv -1$ there (N is odd). Hence $\gcd(\mathsf{X} - \mathsf{Y}, N) = \gcd(\zeta - 1, N) = \prod_{\zeta \equiv 1 \,(p_j^{e_j})} p_j^{e_j}$ with $\zeta = \mathsf{X}/\mathsf{Y}$, which is proper unless $\zeta = \pm 1$. These are 2 of the $2^{\omega(N)}$ elements. □

Remark 6.2 (transcript invariance in Algorithm 3.11)**.** We check that SPLIT satisfies the hypothesis of Theorem 6.1 in each round. Every quantity computed before X_S is a function of the squares R_i and of coins independent of the roots. This covers:

- the field (g, w, c) and its checks;
- the Gauss reduction of Λ_{R_i} (with a fixed tie rule) and the Lenstra–Pomerance factorisation of $|u_0|, |v_0|$, whose coins are fresh;
- the shape decision, and the special-q searches with their cache, which depend on (A, ℓ) only;
- the gcd checks (the unit status of r_i is that of R_i);
- all columns, characters, kernel vectors, the Hensel prime, square-root branches, the choice among three dependencies, and Y_S (Section 10).

The roots enter only through X_S. After a trivial gcd, all rows are discarded and the next round uses fresh roots. For that round we apply Theorem 6.1 with ξ containing everything from earlier rounds, including the old roots, which are independent of the new ones. Field retention biases the law of the field and of the accepted R_i, but not of the fibre signs ϵ_i.

Two cautions. (i) Exactness needs r uniform on all of $[1, N-1]$. Drawing from $[2, N-2]$, for instance, breaks it: at $N = 15$, $R = 1$, the remaining roots $4, 11$ both give proper factors. (ii) Accepted roots are *not* uniform on $(\mathbb{Z}/N)^\times$, because acceptance filters R. What holds, and what is used, is uniformity on the fibre $r_i \boldsymbol{\mu}_2$ given $\mathcal{F}$.

7. Bounded-k splits and field selection

From now on N is the input of SPLIT: odd, with no prime factor $\leq B$ and not a prime power, and $N \geq N_0$.

Lemma 7.1 (reverse Markov)**.** *If $Z \in [0, 1]$ and $\mathbb{E}Z \geq \beta$, then $\Pr[Z \geq \beta/2] \geq \beta/2$.*

Proof. $\beta \leq \mathbb{E}Z \leq \beta/2 + \Pr[Z \geq \beta/2]$. □

Theorem 7.2 (complete descents in one field)**.** *Let A be a field for which* SAMPLE_FIELD *returns A (no rejection and no factor exit), and suppose $|\mathcal{D}_A| \geq 4k^2$. Then one call of*

DESCEND_ONCE(A) *returns a row or makes a factor exit with probability at least*

$$c'_k \Big(\frac{\delta_A}{\log N}\Big)^{2k} 2^{-\omega(N)},$$

with $c'_k > 0$ depending only on k.

Proof. The products of k distinct primes of $\mathcal{D}_A \subset J$ lie in $(Y^k, (2Y)^k] \subset (X2^{-2k}, X]$. By pigeonhole one of the $2k$ dyadic windows $[Y'/2, Y']$ with $Y' = X2^{-i}$, $0 \le i < 2k$, contains a set $\mathcal{S}$ of at least $\binom{|\mathcal{D}_A|}{k}/(2k)$ of them, and $2Y'^2 \le 2X^2 < N$ (Lemma 3.1(e)). Every $\ell \in \mathcal{D}_A$ satisfies $\ell \nmid N$, by definition of N_ℓ. Let E be the event that r is a unit and $R \equiv u/v$ with $u, v \in \mathcal{S}$. Given that r is a unit, R is uniform on Q_N, so Theorem 5.2 gives

$$\Pr_r[E] \ge \frac{\varphi(N)}{N} \cdot \frac{|\mathcal{S}|^2}{c_k h_2 \varphi(N)} = \frac{|\mathcal{S}|^2}{c_k h_2 N}.$$

Now $|\mathcal{S}| \ge (|\mathcal{D}_A|/k)^k/(2k)$ and $|\mathcal{D}_A| = \delta_A \pi_J \ge \delta_A Y/(2\log Y)$, while $Y^{2k} \ge X^2/16^k \ge N/(4\cdot 16^k)$ and $2k \log Y \le \log N$. This gives the displayed lower bound for $\Pr_r[E]$.

On E, Theorem 5.1 shows $(u, v) = j(u_0, v_0)$. Unless a gcd check exits, $|u_0|$ and $|v_0|$ divide u and v, so each is a product of at most k distinct primes of $\mathcal{D}_A$. (This is why the rule says *at most* k: the condition is closed under taking divisors, whereas "exactly k" is not.) So the shape test passes. For each such ℓ we have $N_\ell(A) \ge 1$, and by Proposition 11.2 SQ_SEARCH(A, ℓ) returns an atom unless a gcd check exits first. Hence the call returns a row or exits with a factor. □

Trials and costs. A *trial* is one call of SAMPLE_FIELD followed, if a field is returned, by one call of DESCEND_ONCE. Its outcome is a factor exit, a row, or neither. Let p_F and b be the probabilities of the first two outcomes and $p := p_F + b$. For a field A let ς_A be the probability that DESCEND_ONCE(A) returns a row ($\varsigma_A := 0$ for rejected fields and for fields on which SAMPLE_FIELD exits). Let C_{trial} be a uniform upper bound for the expected cost of one trial and of one call of DESCEND_ONCE, conditional on everything before it. Let T_{post} be a uniform bound for the expected cost of POSTPROCESS. Let $\mathcal{R}$ be the set of fields rejected in steps 2–4 of SAMPLE_FIELD, regardless of exits.

Proposition 7.3 (field selection with factor exits)**.** *(i) If $\gamma \ge L^{-o(1)}$ and $\Pr_{\mathcal{A}}(\mathcal{R}) \le \gamma/2$, then $p \ge L^{-o(1)}$.*

(ii) The expected running time of SPLIT(N) *is at most $3(M_{\text{rows}} C_{\text{trial}} + \mathrm{T}_{\text{post}})/p$.*

Proof. (i) Since $0 \le \delta_A \le 1$, $\mathbb{E}[\delta_A 1_{\mathcal{R}^c}] \ge \gamma - \Pr(\mathcal{R}) \ge \gamma/2$. By Lemma 7.1, the set $\mathcal{G}^* := \{A \notin \mathcal{R} : \delta_A \ge \gamma/4\}$ has probability at least $\gamma/4$. For $A \in \mathcal{G}^*$, either SAMPLE_FIELD exits with a factor, or it returns A. In the latter case $|\mathcal{D}_A| \ge \gamma\pi_J/4 \ge 4k^2$ for large N (as $\pi_J = N^{2\nu+o(1)}$), and Theorem 7.2 applies. Hence $p \ge (\gamma/4)c'_k(\gamma/(4\log N))^{2k} 2^{-\omega(N)} = L^{-o(1)}$, by Lemma 5.4 and because k is fixed.

(ii) Trials are i.i.d., so the number of trials until the outcome is a factor or a row is geometric with mean $1/p$. By Wald's identity the expected cost of selection is at most C_{trial}/p. On a factor exit we are done. Otherwise the outcome is a row, and the kept field A^* has law $\Pr[A^* \in dA] = \varsigma_A \Pr_{\mathcal{A}}(dA)/b$. Given $A^* = A$, a round collects M_{rows} rows with an expected $M_{\text{rows}}/\varsigma_A$ calls (or ends earlier with a factor), at expected cost at most $M_{\text{rows}} C_{\text{trial}}/\varsigma_A + \mathrm{T}_{\text{post}}$. By Theorem 6.1 and Remark 6.2, each round that reaches its final gcd succeeds with conditional probability at least $1/2$, whatever happened before. So the expected cost given $A^* = A$ is at most $2(M_{\text{rows}} C_{\text{trial}}/\varsigma_A + \mathrm{T}_{\text{post}})$. Averaging over the law of A^*,

$$\mathbb{E}\big[1/\varsigma_{A^*}\big] = \int \frac{1}{\varsigma_A} \frac{\varsigma_A}{b} \Pr_{\mathcal{A}}(dA) = \frac{\Pr(\varsigma_A > 0)}{b} \le \frac{1}{b}.$$

The kept-field branch is reached with probability b/p, so its contribution is at most $(b/p) \cdot 2(M_{\text{rows}}C_{\text{trial}}/b + \mathrm{T}_{\text{post}}) \le 2(M_{\text{rows}}C_{\text{trial}} + \mathrm{T}_{\text{post}})/p$. Factor exits during collection only shorten the run. □

Remark 7.4. No lower bound on ς_A for the kept field is used: the harmonic-mean identity makes the cost depend only on b. Nor do we claim that the law of the non-terminating fields retains $\gamma/2$. Factor exits are absorbed as successful outcomes, which repairs a gap in an earlier draft of the argument.

8. The first moment

8.1. **One-point smoothing.** The coefficient vector c enters $F_A(P)$ only through $\langle c, v(P)\rangle$. The next lemma replaces Lee and Venkatesan's Lemmas 5.13, 5.18 and 5.20 [29] by a short self-contained argument.

Lemma 8.1 (one-point smoothing)**.** *Let $P = (a, b)$ with $\gcd(a, b) = 1$ and $ab \neq 0$, put $V' = \max(|a|, |b|)$ and $l_i = |a|^{d-1-i}|b|^i$, and let $H \ge 3dV'$. For c uniform on C_H there is a probability measure ϑ on $\mathbb{Z}$ such that:*

(a) $\|\mathrm{Law}(\langle c, v(P)\rangle) - \vartheta\|_{\mathrm{TV}} \le 3d^2V'/(2H + 1)$;
(b) ϑ is the law of $\sum_i Y_i$ with Y_i independent and uniform on integer intervals of lengths $(2H + 1)l_i$;
(c) $\vartheta(z)$ is a non-increasing function of $|z - M_0|$ for some $M_0 \in \frac{1}{2}\mathbb{Z}$ with $|M_0| \le dV'^{d-1}$;
(d) $\max_z \vartheta(z) \le 1/((2H + 1)V'^{d-1})$.

Proof. Write $v = v(P)$. Its entries have $\gcd(a^{d-1}, b^{d-1}) = 1$. The vectors $u_j := b\,e_j - a\,e_{j+1}$ $(0 \le j \le d-2)$ lie in $\ker\langle\cdot, v\rangle$. The maximal minor of the matrix with rows u_j that omits column i equals $\pm a^{d-1-i}b^i$, so the gcd of the maximal minors is 1, and the u_j are a basis of the lattice $K := \{x \in \mathbb{Z}^d : \langle x, v\rangle = 0\}$. Each coordinate of a point $\sum s_j u_j$ with $s \in [0, 1)^{d-1}$ has absolute value at most $|a| + |b| \le 2V'$.

Representations. Let $m \in \mathbb{Z}$ with $|m| \le \|v\|_2$. Since gcd of the entries is 1, the integer solutions of $\langle x, v\rangle = m$ form a coset $x_0 + K$. The real point $p_m := mv/\|v\|_2^2$ lies on the same affine hyperplane and has $\|p_m\|_\infty \le |m|/\|v\|_2 \le 1$. Writing $p_m - x_0 = \sum s_j u_j$ and $x := x_0 + \sum \lfloor s_j \rfloor u_j$, we get $\langle x, v\rangle = m$ and $\|x\|_\infty \le 1 + 2V'$.

Smoothing. Since $l_i \le V'^{d-1} \le \|v\|_2$, for each i we may fix a set S_i of l_i vectors with $\|x\|_\infty \le 3V'$ on which $\langle\cdot, v\rangle$ is a bijection onto $\{0, \dots, l_i - 1\}$. Let $x^{(i)}$ be uniform on S_i, independent of each other and of c, and put $c' := c + \sum_i x^{(i)}$. Then $\langle c', v\rangle = \sum_i Y_i$ with $Y_i := c_i v_i + \langle x^{(i)}, v\rangle$. Here $c_i v_i = \pm c_i l_i$ with $\pm c_i$ uniform on $[-H, H]$, so Y_i is uniform on an integer interval of length $(2H + 1)l_i$. The Y_i are independent. Let ϑ be the law of $\sum Y_i$. For a fixed shift y, the uniform measures on C_H and $C_H + y$ differ in total variation by at most $\|y\|_1/(2H + 1)$. Hence $\|\mathrm{Law}(c) - \mathrm{Law}(c')\|_{\mathrm{TV}} \le d \cdot 3dV'/(2H + 1)$, and (a) follows by applying the map $\langle\cdot, v\rangle$.

(c) A uniform distribution on an integer interval is a log-concave sequence, symmetric about the midpoint of the interval. Convolutions of log-concave sequences without internal zeros are log-concave (see e.g. [49]), and a log-concave sequence is unimodal. A symmetric unimodal sequence is non-increasing in the distance to its centre. The centre of Y_i is $(l_i - 1)/2$ or its negative, so $|M_0| \le \sum l_i/2 \le dV'^{d-1}$. (d) A convolution of probability measures has maximum at most the maximum of any factor. Take the factor with $l_i = V'^{d-1}$. □

By (c), ϑ has a layer-cake decomposition

$$\vartheta = \sum_m W_m \,\mathrm{Unif}\big([M_0 - y_m, M_0 + y_m] \cap \mathbb{Z}\big), \qquad W_m \ge 0, \quad \sum_m W_m = 1, \tag{5}$$

because every level set $\{z : \vartheta(z) \ge s\}$ is an integer interval symmetric about M_0.

8.2. **Good moduli.** Put $\varpi := L$ and

$$\mathcal{X} := \big\{X^*\varpi^{-2}(3/2)^i : i \ge 0\big\} \cap \big[X^*\varpi^{-2},\, 16dX^*\big], \qquad |\mathcal{X}| = O(\mathcal{L}).$$

The grid $\mathcal{X}$ does not depend on ℓ, P, g, w, or any residue.

Definition 8.2. A positive integer t is *good* if, for every $x \in \mathcal{X}$ and every r with $(r,t) = 1$, $|\Psi(x;t,r) - \Psi_t(x)/\varphi(t)| \le \Psi_t(x)/(2\varphi(t))$. Otherwise t is *bad*.

Lemma 8.3 (algebraic smoothness at a good modulus)**.** *Let $N \ge N_0$. Let $\ell \in J$, $P = (a,b) \in \mathcal{P}_\ell$, $g \in \mathcal{G}$ and w be such that $t := |G(P)| \in [MV_\ell/32, 2MV_\ell]$, t is good, and $r_0 := \hat{F}_{g,w}(P) \bmod t$ satisfies $(r_0, t) = 1$. Then*

$$\Pr_c\big[F_A(P) \in \mathcal{S}_B\big] \;\ge\; \rho_F/(\log N)^C.$$

Proof. Write $V = V_\ell$, $V' = \max(|a|, b) \in (V/2, V]$, $\hat{F} = \hat{F}_{g,w}(P)$ and $G = G(P)$. Since $F_A(P) = \hat{F} + G\langle c, v(P)\rangle$ and $H \ge 3dV'$, Lemma 8.1 gives $\Pr_c[F_A(P) \in \mathcal{S}_B] \ge \Pr_{Z\sim\vartheta}[\hat{F} + GZ \in \mathcal{S}_B] - 3d^2V/(2H+1)$. Use (5) and put $Y_\varpi := (2H+1)V'^{d-1}/\varpi$.

Small layers. The layers with $y_m < Y_\varpi$ are supported in $|z - M_0| < Y_\varpi$, a set of ϑ-mass at most $(2Y_\varpi + 1)/((2H+1)V'^{d-1}) \le 3/\varpi$ by Lemma 8.1(d).

One large layer. Fix m with $y := y_m \ge Y_\varpi$ and put $y' := \lfloor y/2 \rfloor$. Since $|M_0| \le dV'^{d-1} \le Y_\varpi/2$, the layer's interval contains $[0, y']$ and $[-y', 0]$. Consider $z \in [0, y']$. The values $\hat{F} + Gz$ are all integers $\equiv r_0 \pmod t$ between $\hat{F}$ and $\hat{F} + Gy'$. If $G > 0$ they include every $n \equiv r_0$ with $|\hat{F}| < n \le ty' - |\hat{F}|$. If $G < 0$ they include every $n \equiv r_0$ with $-(ty' - |\hat{F}|) \le n < -|\hat{F}|$, that is, every $-n \equiv -r_0$ in the same range of $|n|$. So, with $r' = \pm r_0$ (a reduced residue),

$$\#\{0 \le z \le y' : \hat{F} + Gz \in \mathcal{S}_B\} \;\ge\; \Psi(x_1; t, r') - \Psi(|\hat{F}|; t, r')$$

for any $x_1 \le ty' - |\hat{F}|$. By Lemma 3.5, $|\hat{F}| \le C_d MHV^d N^{-\eta_0/2}$, while $ty' \ge (MV/32)(Y_\varpi/2 - 1) \ge c_d MHV^d/\varpi$. So $|\hat{F}| \le ty'/4$, and also $ty'/4 \ge X^*\varpi^{-2}$ and $ty'/2 \le 2MV \cdot d(H+1)V^{d-1} \le 16dX^*$. Consecutive points of $\mathcal{X}$ differ by the factor $3/2 < 2$, so we may take $x_1 \in \mathcal{X} \cap (ty'/4, ty'/2]$. Since t is good, and by Fact 2.3 (here $\omega(t) \le \log N$, so its error term is $o(1)$),

$$\Psi(x_1; t, r') \ge \frac{\Psi_t(x_1)}{2\varphi(t)} \ge \frac{\Psi(x_1)}{3t} \ge \frac{y'}{12}\varrho(ty'/4) \ge \frac{y'\rho_F}{(\log N)^C},$$

using Lemma 2.4(i),(ii): $ty'/4$ and X^* differ by a factor $\le \varpi^2 \le B^C$. Trivially $\Psi(|\hat{F}|; t, r') \le |\hat{F}|/t + 1$. The same bound holds for $z \in [-y', 0]$. Dividing by the $2y + 1$ points of the layer, the proportion of smooth values in the layer is at least

$$\frac{\rho_F}{3(\log N)^C} - \frac{|\hat{F}|/t + 1}{y} \;\ge\; \frac{\rho_F}{3(\log N)^C} - C\varpi N^{-\eta_0/2}.$$

Conclusion. $\Pr_c \ge (1 - 3/\varpi)\big(\rho_F/(3(\log N)^C) - C\varpi N^{-\eta_0/2}\big) - 3d^2V/(2H+1)$. Here $\rho_F = L^{-O(1)}$, $\varpi = L$ and $V/H \le N^{-\nu-\eta_0}$, so the error terms are negligible. □

8.3. **Bad moduli are rare.** The next lemma is a restricted Bombieri–Vinogradov estimate for moduli of the form $\ell \cdot$(smooth). It is the analogue of Lee–Venkatesan's Lemma 8.5 [29] for these moduli. Its proof, in Appendix A, does not use that lemma; it rests on the exact identity of Lemma A.8, the large sieve, and Drappeau's Lemme 5.

Lemma 8.4 (restricted BV for the moduli ℓs; "Lemma 8.5′")**.** *Fix $\xi > 0$ and q_0 with $2\nu < q_0 < \xi/2$. Let $x = N^{\xi+o(1)}$ and $Q = N^{q_0+o(1)}$, and let*

$$\mathcal{T}_Q := \{t = \ell s \in (Q/2, Q] : \ \ell \in J, \ s \geq 1, \ P^+(s) \leq B\}.$$

Then, uniformly in such x, Q,

$$\sum_{t \in \mathcal{T}_Q} \max_{(r,t)=1} \Big| \Psi(x;t,r) - \frac{\Psi_t(x)}{\varphi(t)} \Big| \ \leq \ \varepsilon_N \, \Psi(x) \, \frac{\#\mathcal{T}_Q}{Q},$$

where $\varepsilon_N = C\exp\big(-c_0(\log N)^{1/2}(\log\log N)^{-5/2}\big)$ for some fixed $C, c_0 > 0$. In particular $\varepsilon_N(\log N)^A \to 0$ for every fixed A.

Corollary 8.5. *In the setting of Lemma 8.4 with x ranging over $\mathcal{X}$, the number of bad $t \in \mathcal{T}_Q$ is at most $3|\mathcal{X}|\varepsilon_N \#\mathcal{T}_Q$.*

Proof. If $t \in \mathcal{T}_Q$ is bad, there are $x \in \mathcal{X}$ and $(r,t) = 1$ with $|\Psi(x;t,r) - \Psi_t(x)/\varphi(t)| > \Psi_t(x)/(2\varphi(t)) \geq \Psi(x)/(3t) \geq \Psi(x)/(3Q)$, by Fact 2.3. Sum Lemma 8.4 over $x \in \mathcal{X}$ and apply Markov's inequality. □

8.4. **Rational fibres.**

Lemma 8.6 (fibres)**.** *Fix $\ell \in J$ and $P = (a,b) \in \mathcal{P}_\ell$, and put $\mathcal{G}_P(n) := \{g \in \mathcal{G} : G_g(P) = \ell n\}$.*

(i) The $g \in \mathbb{Z}^2$ with $G_g(P) = \ell n$ form a line $g^* + \mathbb{Z}(a,b)$, and $|\mathcal{G}_P(n)| \leq \sqrt{2}M/V_\ell + 1 \leq 2M/V_\ell$ for every n.

(ii) There is a set I_P of consecutive integers n, of length at least $MV_\ell/(16\ell) - 1$, on which ℓn has constant sign, $MV_\ell/32 \leq \ell|n| \leq 2MV_\ell$, and $|\mathcal{G}_P(n)| \geq M/(CV_\ell \log\log N)$.

Proof. Write $V = V_\ell$. (i) $G_g(P) = g_1 a - g_0 b$ is constant along the direction (a,b), which is primitive with $|(a,b)| \geq b > V/2$, and the square $[M/2, M]^2$ has diameter $M/\sqrt{2}$.

(ii) Let (g_0, g_1) be uniform on the real square $[M/2, M]^2$. Then $G = ag_1 - bg_0$ is a sum of independent uniform variables on intervals of lengths $\alpha \leq \beta$ with $\beta = V'M/2$, $V' = \max(|a|, b)$. Its density is at least $1/(2\beta)$ on an interval of length β. The density at s equals (length of the chord $\{G = s\}$)$/(|(a,b)|M^2/4)$, so on that interval the chord has length at least $(1/(V'M)) \cdot V' \cdot M^2/4 = M/4$. Removing $(-MV/32, MV/32)$ from an interval of length $\geq MV/4$ leaves a subinterval of length $\geq MV/16$ of constant sign. Let I_P be the integers n with ℓn in it. For $n \in I_P$ the fibre line meets the closed square in at least $\lfloor (M/4)/|(a,b)| \rfloor \geq M/(8V)$ consecutive lattice points $g^* + j(a,b)$.

Coprimality. Fix $e \geq 1$. Choose x, y with $xa + yb = 1$. If $e \mid \gcd(g_0, g_1)$ then $e \mid G = \ell n$. Conversely, if $e \mid \ell n$, the j with $e \mid g^* + j(a,b)$ are exactly those with $j \equiv -(xg_0^* + yg_1^*) \pmod e$: the matrix $\left(\begin{smallmatrix} x & y \\ -b & a \end{smallmatrix}\right)$ has determinant 1 and maps g to $(xg_0 + yg_1, G)$. By Möbius inversion over $e \mid \ell|n|$, the number of coprime points among $J_0 \geq M/(8V)$ consecutive ones is at least $J_0\varphi(\ell|n|)/(\ell|n|) - 2^{\omega(\ell|n|)}$. Finally $\varphi(m)/m \gg 1/\log\log m$, and $2^{\omega(m)} = N^{o(1)}$ is small compared with $M/V \geq N^{\eta_0/2}$ (Lemma 3.1(c)). □

8.5. **The first moment.**

Theorem 8.7 (first moment)**.** *For $N \geq N_0$,*

$$\mathbb{E}_{\ell \in J} \mathbb{E}_{A \sim \mathcal{A}} N_\ell(A) \ \geq \ \frac{S^2 \rho_r \rho_F}{(\log N)^C}.$$

Proof. At most $O(1)$ primes of J divide N, and we drop them. For $\ell \nmid N$,

$$\mathbb{E}_A N_\ell(A) \;\geq\; \frac{1}{|\mathcal{G}|} \sum_{P \in \mathcal{P}_\ell} \sum_{\substack{n \in I_P,\ n \in \mathcal{S}_B \\ \ell \mid n| \text{ good}}} \sum_{g \in \mathcal{G}_P(n)} \mathbb{E}_w \Pr_c\big[F_A(P) \in \mathcal{S}_B\big].$$

This is an inequality, because the filter $|G(P)| \geq MV_\ell/32$ holds on I_P and the other n are dropped. For these (ℓ, P, n, g) and every w, Lemma 3.8 shows that r_0 is reduced, so Lemma 8.3 bounds the inner probability below by $\rho_F/(\log N)^C$. Hence

$$\mathbb{E}_A N_\ell(A) \geq \frac{\rho_F}{(\log N)^C |\mathcal{G}|} \sum_{P \in \mathcal{P}_\ell} \Big(\sum_{n \in I_P \cap \mathcal{S}_B} |\mathcal{G}_P(n)| - \frac{2M}{V_\ell} \beta_\ell \Big),$$

where $\beta_\ell := \#\{n : 1 \leq |n| \leq 2MV_\ell/\ell,\ n \in \mathcal{S}_B,\ \ell|n| \text{ bad}\}$ depends on ℓ only.

Good mass. I_P is an interval of length $\lambda \geq MV_\ell/(16\ell) - 1$ whose right end (in absolute value) is at most 32λ. By Lemma 2.4(iii),(i) it contains at least $c\lambda\rho_r$ smooth integers. By Lemma 8.6(ii), $\sum_{n \in I_P \cap \mathcal{S}_B} |\mathcal{G}_P(n)| \geq cM^2\rho_r/(\ell \log\log N)$.

Bad mass. The moduli $\ell|n|$ counted by the β_ℓ lie in $\bigcup_Q \mathcal{T}_Q$ for $O(1)$ dyadic values Q covering $[MV_-/32, 2MV_{\max}]$, all with $Q = N^{\mu+\nu+o(1)}$. Lemma 8.4 applies with $\xi = \mu + h + d\nu$ and $q_0 = \mu + \nu$, because $q_0 - 2\nu = \mu - \nu \geq \eta_0$ and $\xi/2 - q_0 = (h - \mu + (d-2)\nu)/2 \geq \eta_0/2$ by the level floor. By Corollary 8.5,

$$\frac{1}{\pi_J} \sum_{\ell \in J} \beta_\ell \leq \frac{2}{\pi_J} \sum_Q 3|\mathcal{X}|\varepsilon_N \#\mathcal{T}_Q \leq C|\mathcal{X}|\varepsilon_N \frac{MV_{\max}}{Y} \rho_r (\log N)^C,$$

since $\#\mathcal{T}_Q \leq \sum_{\ell \in J} \Psi(Q/\ell) \leq C\pi_J(MV_{\max}/Y)\rho_r$ by Lemma 2.4(i).

Assembly. $\#\mathcal{P}_\ell \asymp V_\ell^2/\log V_\ell$ (there are $\asymp V/\log V$ primes $b \in (V/2, V]$, and for each at least V values of a coprime to b), $|\mathcal{G}| \asymp M^2$, and $V_\ell^2/\ell \geq S^2/2$. Averaging over ℓ, the good mass contributes at least $cS^2\rho_r\rho_F/(\log N)^{C'}$. The bad mass contributes at most $CS^2\rho_r\rho_F|\mathcal{X}|\varepsilon_N(\log N)^{C''}$, which is smaller by the factor $|\mathcal{X}|\varepsilon_N(\log N)^{O(1)} \to 0$. □

Remark 8.8. The bound is averaged over $\ell \in J$ as well as over the field. The bad-modulus step does not give it for each individual ℓ, and no such statement is needed. The loss is polylogarithmic: $1/\log V$ from prime b, $\log\log N$ from $\varphi(m)/m$, and the $(\log N)^{O(1)}$ of Lemma 2.4(ii).

9. The second moment and relation supply

Throughout this section $\ell \in J$ with $\ell \nmid N$, $V = V_\ell$, and $T = T_\ell = dHV^{d-1}$. For $P \in \mathcal{P}_\ell \cap \Lambda_\ell(g)$ write $Y_P = Y_P(g) := 1$ if $|G_g(P)| \geq MV/32$ and $G_g(P)/\ell \in \mathcal{S}_B$, and $Y_P := 0$ otherwise.

9.1. Pairs in one lattice.

Lemma 9.1 (pairs)**.** *Let $P \neq P'$ in $\mathcal{P}_\ell \cap \Lambda_\ell(g)$ and $\Delta := \det(P, P')$. Then $\Delta \neq 0$ and $\ell \mid \Delta$. Put $q := |\Delta|/\ell$; then $q \leq 2S^2$. For fixed P and q, at most 12 points $P' \in \mathcal{P}_\ell$ satisfy $|\det(P, P')| = \ell q$.*

Proof. Both points have $b, b' > 0$ and are primitive, so they are not proportional and $\Delta \neq 0$. They are both annihilated modulo ℓ by the nonzero linear form $(x, y) \mapsto g_1 x - g_0 y$, so they are proportional modulo ℓ and $\ell \mid \Delta$. Also $|\Delta| \leq 2V^2 \leq 2\ell S^2$. For fixed P and D, the solutions of $\det(P, P') = D$ form a line $P'' + \mathbb{Z}P$. It meets $[-V, V]^2$ in a segment of length at most $2\sqrt{2}V$, and consecutive points differ by $|P| \geq V/2$. That gives at most 6 points for each of $D = \pm\ell q$. □

Lemma 9.2 (kernel transfer)**.** *Let $d \geq 3$ and let $P = (a, b) \neq P' = (a', b')$ be primitive, with $\Delta = \det(P, P') \neq 0$. Put $v = v(P)$, $v' = v(P')$ and $\Phi : \mathbb{Z}^d \to \mathbb{Z}^2$, $c \mapsto (\langle c, v\rangle, \langle c, v'\rangle)$. Let κ be an integer with $P' \equiv \kappa P \pmod{\Delta}$.*

(i) $\Gamma := \Phi(\mathbb{Z}^d) = \{(X, X') \in \mathbb{Z}^2 : X' \equiv \kappa^{d-1} X \pmod{\Delta}\}$, *of index* $|\Delta|$, *and* κ *is a unit modulo* Δ.

(ii) $\ker \Phi = Q_{PP'} \cdot \mathbb{Z}[x, y]_{d-3}$ *with* $Q_{PP'} = (bx - ay)(b'x - a'y)$, *identifying* $\mathbb{Z}^d$ *with forms of degree* $d - 1$.

(iii) If $|a|, |b|, |a'|, |b'| \le V$ *and* $H \ge 4V^2$, *then for every* $z \in \mathbb{Z}^2$, $\Pr_{c \in C_H}[\Phi(c) = z] \le C_d |\Delta| / (H^2 \|v \wedge v'\|) \le C_d / (H^2 (bb')^{d-2})$.

Proof. For $\gamma \in GL_2(\mathbb{Z})$, $v(\gamma P) = \rho(\gamma) v(P)$ with $\rho(\gamma) \in GL_d(\mathbb{Z})$ (the symmetric power). By Cauchy–Binet, the gcd of the 2×2 minors of the matrix with rows v, v' is unchanged. Choose γ with $\gamma P = (1, 0)$. Then $\gamma P' = (u, \pm\Delta)$ with $\gcd(u, \Delta) = 1$, and the nonzero minors of the matrix with rows $(1, 0, \dots, 0)$ and $v(u, \pm\Delta)$ are $\pm u^{d-1-j} \Delta^j$, $j \ge 1$. Their gcd is $|\Delta|$. Hence $\Phi(\mathbb{Z}^d)$ has index $|\Delta|$. In these coordinates $P' \equiv uP \pmod{\Delta}$, so κ exists, is a unit, and by homogeneity $v' \equiv \kappa^{d-1} v \pmod{\Delta}$. Thus $\Phi(\mathbb{Z}^d)$ lies in the displayed lattice, which also has index $|\Delta|$. This proves (i).

(ii) $\langle c, v(P) \rangle = C(P)$ for the form $C = \sum c_i x^{d-1-i} y^i$. If $C(P) = C(P') = 0$ then $(bx - ay) \mid C$ and $(b'x - a'y) \mid C$ over $\mathbb{Q}$, and the two factors are coprime since $\Delta \neq 0$. By Gauss's lemma the quotient is integral. Conversely every such multiple lies in the kernel.

(iii) The kernel K_2 has rank $d - 2$. It is saturated, and its orthogonal complement in $\mathbb{Z}^d$ is the saturation of $\mathbb{Z}v + \mathbb{Z}v'$. That lattice has determinant $\|v \wedge v'\|$ and index $|\Delta|$ in its saturation (the gcd of the minors, by (i)). A saturated lattice and its orthogonal complement have the same determinant, so $\det K_2 = \|v \wedge v'\| / |\Delta|$. The monomial multiples of $Q_{PP'}$ form a basis of K_2 with entries at most $2V^2$, so a fundamental cell has ℓ^∞-diameter at most $2(d-2)V^2 \le dH$. A fibre $\Phi^{-1}(z) \cap C_H$ is $(c_0 + K_2) \cap C_H$. The translates of the cell at its points are disjoint and lie in the intersection of the cube $[-(d+1)H, (d+1)H]^d$ with a $(d-2)$-flat, whose $(d-2)$-volume is at most $C_d H^{d-2}$. Hence the fibre has at most $C_d H^{d-2} |\Delta| / \|v \wedge v'\|$ points, and we divide by $(2H+1)^d$. Finally, the minor of columns $d-2, d-1$ equals $ab^{d-2}b'^{d-1} - b^{d-1}a'b'^{d-2} = (bb')^{d-2}\Delta$, so $\|v \wedge v'\| \ge |\Delta| (bb')^{d-2}$. □

For candidates $b, b' > V/2$, so (iii) gives $\Pr_c[\Phi(c) = z] \le C_d / (H^2 V^{2d-4})$. The kernel floor $h \ge 2\nu + \eta_0$ gives $H \ge 4V^2$ for large N.

9.2. **The multiplier.**

Lemma 9.3 (residue spreading)**.** *For* $N \ge N_0$, *every* $1 \le t \le 2MV_{\max}$ *and every* $r \in \mathbb{Z}$,

$$\Pr_w[w \equiv r \pmod{t}] \le \frac{C_W}{\varphi(t)}, \qquad C_W := \frac{24}{\eta_0}.$$

Consequently, if (ℓ, P, g) *satisfy the hypotheses of Lemma 3.8 and* $\Phi_0 : \mathbb{Z}/t \to [0, \infty)$ *does not depend on* w, *then* $\mathbb{E}_w \Phi_0(\hat{F}_{g,w}(P) \bmod t) \le (C_W/\varphi(t)) \sum_{c \in (\mathbb{Z}/t)^\times} \Phi_0(c)$.

Proof. If $(r, t) > 1$ the probability is 0, since every w is a prime larger than t. Otherwise Fact 2.8 with $x = W$, $y = W$ gives $\pi(2W; t, r) - \pi(W; t, r) < 2W/(\varphi(t) \log(W/t))$, while $\pi(2W) - \pi(W) \ge W/(2 \log W)$. So the probability is at most $4 \log W / (\varphi(t) \log(W/t))$. Here $\log W \le \log N$, and $\log(W/t) \ge (\eta_0/3) \log N$ by Lemma 3.1(b). For the consequence, $\hat{F}_{g,w}(P) \equiv c_P w$ with c_P a unit independent of w (Lemma 3.8), and $x \mapsto c_P x$ permutes $(\mathbb{Z}/t)^\times$. □

9.3. **Divisor switching.**

Lemma 9.4 (smooth numbers in short progressions)**.** *Fix* $C_0 \ge 1$. *Let* X *be in range,* $1 \le q \le B^{C_0}$, *and let* $I \subset [XB^{-C_0}, X]$ *be an interval of length* $A \ge XB^{-C_0}$. *Then for every* $r \in \mathbb{Z}$,

$$\#\{n \in I : \ n \equiv r \pmod{q}, \ P^+(n) \le B\} \ \le \ \frac{A}{q} \varrho(X) (\log N)^{O(C_0)}.$$

Proof. Let $e = \gcd(r, q)$. If e has a prime factor $> B$ the count is 0. Otherwise write $n = en_1$. Then n_1 lies in I/e, an interval of length A/e, and $n_1 \equiv r/e \pmod{q'}$ with $q' = q/e$ and $\gcd(r/e, q') = 1$, so $(n_1, q') = 1$. Put $D := \min(A/q, X/(eB^{C_0}))$. Then $D \le \min(I/e)$. Every smooth $n_1 \ge D$ has a divisor $d_1 \in (D/B, D]$: take the largest product of its smallest prime factors not exceeding D. Write $n_1 = d_1 m$. Then $(d_1, q') = 1$, $m \equiv (r/e)d_1^{-1} \pmod{q'}$, and m lies in an interval of length $A/(ed_1)$. So there are at most $A/(d_1 q) + 1 \le 2A/(d_1 q)$ choices of m (as $d_1 \le A/q$), and the count is at most $(2A/q)\sum_{D/B < d_1 \le D,\, P^+(d_1) \le B} 1/d_1$. Splitting into dyadic blocks and using Lemma 2.4(i), this sum is at most $8\log_2(2B)\varrho(D/B)$. Finally $D \ge XB^{-2C_0}$, so $X/(D/B) \le B^{2C_0+1}$, and Lemma 2.4(ii) gives $\varrho(D/B) \le \varrho(X)(\log N)^{O(C_0)}$. □

9.4. **An all-residue second moment for the family $\ell^2 nq$.** Put $\mathcal{S}^* := \mathcal{S}_B \cap [-X^*, X^*]$, so $\#\mathcal{S}^* \le 2\Psi(X^*)$, and

$$S(\alpha) := \sum_{F \in \mathcal{S}^*} e(\alpha F), \qquad \mathfrak{Q}(m) := \#\{(F, F') \in \mathcal{S}^{*2} : F \equiv F' \pmod{m}\},$$

$$f(m') := \sum_{\substack{b \bmod m' \\ (b, m') = 1}} |S(b/m')|^2.$$

Orthogonality and the reduction of fractions give the exact identity

$$\mathfrak{Q}(m) = \frac{1}{m} \sum_{h \bmod m} |S(h/m)|^2 = \frac{1}{m} \sum_{m' \mid m} f(m'), \qquad f(1) = (\#\mathcal{S}^*)^2. \tag{6}$$

Let $X'_+ := 2MV_{\max}/Y = X_r$, and $\mathcal{N}_\ell := \{n \in \mathbb{Z} : \ n \in \mathcal{S}_B,\ MV_\ell/32 \le \ell|n| \le 2MV_\ell\}$, so $\#\mathcal{N}_\ell \le 2\Psi(X'_+)$. Fix C_0 so that

$$R_0 := B^{C_0} \ \ge \ 64S^2 L/(\rho_r \rho_F)$$

for large N. This is possible because the right side is $L^{O(1)}$. Note that $R_0 < Y$.

Theorem 9.5 (L^2 bound for the family; "$L2_{\text{fam}}$")**.** *Uniformly for $1 \le q \le 2S^2$,*

$$\mathbb{E}_{\ell \in J} \sum_{n \in \mathcal{N}_\ell} \sum_{m' \mid \ell^2 |n| q,\ m' > 1} f(m') \ \le \ \tau(q)(\log N)^C \, \Psi(X'_+)\Psi(X^*)^2, \tag{7}$$

and consequently

$$\mathbb{E}_{\ell \in J} \sum_{n \in \mathcal{N}_\ell} \frac{\mathfrak{Q}(\ell^2 |n| q)}{\varphi(\ell |n|)} \ \le \ \tau(q)(\log N)^C \, \frac{\Psi(X'_+)\Psi(X^*)^2}{Yq(MV_-)^2}. \tag{8}$$

Proof. Every prime factor of $|n|q$ is at most $\max(B, 2S^2) < Y$, so $|n|q$ has no prime factor in J.

Rearrangement. Since X^* does not depend on ℓ, neither does f. So the left side of (7) is $\sum_{m' > 1} f(m')\nu(m')$ with $\nu(m') := \mathbb{E}_\ell \#\{n \in \mathcal{N}_\ell : m' \mid \ell^2 |n| q\}$. Partition the $m' > 1$ into three classes: (A) no prime factor in J and $m' \le R_0$; (B) no prime factor in J and $m' > R_0$; (C) some prime factor in J. The partition does not depend on ℓ. In classes (A) and (B), $m' \mid \ell^2 |n| q$ iff $m' \mid |n| q$. In class (C), $m' \mid \ell^2 |n| q$ forces the J-primes of m' to be exactly $\{\ell\}$, with exponent $i \in \{1, 2\}$.

For any m' write $m_2 := \gcd(m', q)$ and $m_1 := m'/m_2$. Then $m' \mid |n|q$ iff $m_1 \mid n$, because $\gcd(m_1, q/m_2) = 1$. The map $m' \mapsto (m_1, m_2)$ is injective and $m_2 \mid q$.

Class (A). By Parseval over all $b \bmod m'$, $f(m') \le m' \sum_{c \bmod m'} N_c^2$ with $N_c := \#\{F \in \mathcal{S}^* : F \equiv c \ (m')\}$. Lemma 9.4 on $\pm[X^* R_0^{-1}, X^*]$, plus the trivial bound $X^*/(R_0 m') + 1$ on $|F| < X^*/R_0$, gives $N_c \le (X^*/m')\rho_F (\log N)^C$ (using $R_0^{-1} \le \rho_F$ and $m' \le R_0 \le X^* \rho_F$). Hence $f(m') \le m' \max_c N_c \cdot \#\mathcal{S}^* \le \Psi(X^*)^2 (\log N)^C$. For the multiplicity, $\nu(m') = 0$ unless m_1 is

smooth, and then $\nu(m') \le 2\Psi(X'_+/m_1) \le (\Psi(X'_+)/m_1)(\log N)^{O(C_0)}$ by Lemma 2.4(ii). Since $\sum_{m' \le R_0} 1/m_1 \le \tau(q)(1+\log R_0)$, class (A) contributes at most $\tau(q)(\log N)^C \Psi(X'_+)\Psi(X^*)^2$.

Class (B). Here $\nu(m') \le \#\{1 \le |n| \le X'_+ : m_1 \mid n\} \le 2X'_+q/m'$, and $\nu(m') = 0$ unless $m' \le X'_+q$. On dyadic blocks $(R, 2R]$ with $R = R_0 2^i \le X'_+q$, Fact 2.7 with $K = 2\lfloor X^* \rfloor + 1$ gives $\sum_{m' \in (R,2R]} f(m') \le (3X^* + 4R^2) \cdot 2\Psi(X^*)$. Summing over blocks, class (B) contributes at most $CX'_+qX^*\Psi(X^*)/R_0 + C(X'_+q)^2\Psi(X^*)$. Relative to $\Psi(X'_+)\Psi(X^*)^2 = X'_+X^*\rho_r\rho_F\Psi(X^*)$ these are $Cq/(R_0\rho_r\rho_F) \le 1$ and $CX'_+q^2/(X^*\rho_r\rho_F) = N^{-h-(d+1)\nu}L^{O(1)} = o(1)$.

Class (C). Let $\ell = \ell(m')$ be the unique J-prime of m' and $m'' := m'/\ell^i$. Only this ℓ contributes, so $\nu(m') = \pi_J^{-1}\#\{n \in \mathcal{N}_\ell : m'' \mid |n|q\}$. We have two bounds: $\nu(m') \le 2\Psi(X'_+)/\pi_J$, and, as $m'' \ge m'/(4Y^2)$, $\nu(m') \le 8X'_+qY^2/(\pi_J m')$. The relevant m' lie in $[Y, 4Y^2X'_+q]$, which is $O(\log N)$ dyadic blocks. On each block use the large sieve bound $(3X^* + 4R^2) \cdot 2\Psi(X^*)$, and the first bound for ν against $3X^*$ and the second against $4R^2$ (if $\nu \le A_1$ and $\nu \le A_2$ then $\nu(K_1+K_2) \le A_1K_1 + A_2K_2$). Class (C) contributes at most

$$\frac{C\log N}{\pi_J}\Psi(X'_+)X^*\Psi(X^*) + \frac{C}{\pi_J}(X'_+)^2q^2Y^4\Psi(X^*).$$

Relative to the target these are $C\log N/(\pi_J\rho_F) = N^{-2\nu}L^{O(1)}$ and $CX'_+q^2Y^4/(\pi_JX^*\rho_r\rho_F) = N^{(5-d)\nu-h}L^{O(1)}$. The latter is $o(1)$ because $h \ge 2\nu + \eta_0 > (5-d)\nu$ for every $d \ge 3$. This proves (7).

(8). For $n \in \mathcal{N}_\ell$ put $m = \ell^2|n|q = \ell qt$ with $t = \ell|n| \ge MV_-/32$, and note $\varphi(t) \ge t/(C\log\log N)$. By (6), $\mathfrak{Q}(m)/\varphi(t) \le C\log\log N\,\big[(\#\mathcal{S}^*)^2 + \sum_{m'|m, m'>1} f(m')\big]/(Yq(MV_-)^2)$. Sum over n, average over ℓ, and use $\#\mathcal{N}_\ell \le 2\Psi(X'_+)$ and (7). □

Remark 9.6. The average over ℓ is essential in class (C). For a single ℓ, the frequencies with denominator ℓ^2 contribute $\ell^2\sum_c N_c^2$ over classes modulo ℓ^2, whose trivial bound is $X^*\Psi(X^*) \gg \Psi(X^*)^2$. Improving it would require L^2-equidistribution of smooth numbers modulo $\ell^2 = B^{\Theta(u)}$, which is out of reach unconditionally. Note also that the statement is an upper bound at the main-term scale, with no saving; it does not follow from Harper's Barban–Davenport–Halberstam theorem over all moduli [18, Thm. 2], whose error term $\Psi^2L^{-o(1)}$ exceeds the family's main term.

9.5. **The second moment.**

Theorem 9.7 (second moment; "LD*")**.** *For $N \ge N_0$,*

$$\mathbb{E}_{\ell\in J}\mathbb{E}_{A\sim\mathcal{A}}\big[N_\ell(A)(N_\ell(A)-1)\big] \;\le\; (\log N)^C\, S^4\rho_r^2\rho_F^2.$$

Proof. Step 0 (law). If $\ell \mid N$ then $N_\ell = 0$. For $\ell \nmid N$, $N_\ell(N_\ell - 1)$ counts ordered pairs of distinct counted points, so

$$\mathbb{E}_A[N_\ell(N_\ell-1)] \le \mathbb{E}_{g,w}\sum_{P\ne P'} Y_PY_{P'}\Pr_c\big[F_A(P), F_A(P') \in \mathcal{S}_B\big],$$

the sum running over $P \ne P'$ in $\mathcal{P}_\ell \cap \Lambda_\ell(g)$. By Lemma 9.1, $|\Delta| = \ell q$ with $1 \le q \le 2S^2$. Only $\hat{F}_{g,w}$ depends on w.

Step 1 (kernel). By Lemma 9.2(iii), $\Pr_c[\Phi(c) = z] \le C_d/(H^2V^{2d-4})$. Since $\Phi(C_H) \subset [-T,T]^2$ and $\Gamma = \{z_2 \equiv \kappa^{d-1}z_1\ (\Delta)\}$,

$$\Pr_c[\text{both smooth}] \le \frac{C_d}{H^2V^{2d-4}}\,Z, \qquad Z := \sum_{j \bmod |\Delta|} A_j(P)A_{\kappa^{d-1}j}(P'),$$

where $A_j(P) := \#\{|z| \le T : z \equiv j\ (|\Delta|),\ \hat{F}(P) + G(P)z \in \mathcal{S}_B\}$.

Step 2 (AM–GM). κ is a unit modulo Δ, so $j \mapsto \kappa^{d-1} j$ permutes the classes and $Z \le \frac{1}{2}(\mathfrak{q}(P) + \mathfrak{q}(P'))$ with $\mathfrak{q}(P) := \sum_j A_j(P)^2$. This inequality is pointwise but one-sided.

Step 3 (enlargement). $z \mapsto F = \hat{F}(P) + G(P)z$ is injective, $F \equiv r_0 \pmod t$ with $t = |G(P)|$, and $z \equiv z' \pmod{|\Delta|}$ iff $F \equiv F' \pmod{t|\Delta|}$, for either sign of $G(P)$. All such F lie in $[-X^*, X^*]$ (Lemma 3.5). Hence

$$\mathfrak{q}(P) \le \mathfrak{Q}_0(t, |\Delta|; r_0) := \#\{(F, F') \in \mathcal{S}^{*2} : F \equiv F' \equiv r_0 \ (t),\ F \equiv F' \ (t|\Delta|)\},$$

a function of r_0 alone, which does not depend on w otherwise.

Step 4 (average over w). When $Y_P = 1$, Lemma 9.3 applies to $\Phi_0 = \mathfrak{Q}_0(t, |\Delta|; \cdot)$. Summing over the reduced classes counts a subset of the pairs with $F \equiv F' \pmod{t|\Delta|}$, so $\mathbb{E}_w \mathfrak{Q}_0(t, |\Delta|; r_0) \le (C_W/\varphi(t))\mathfrak{Q}(t|\Delta|)$. The same holds for P'. The pair sum is symmetric under $(P, P') \mapsto (P', P)$, so

$$\mathbb{E}_A[N_\ell(N_\ell - 1)] \le \frac{C_d C_W}{H^2 V_-^{2d-4}}\, \mathbb{E}_g \sum_{P \ne P'} Y_P Y_{P'} \frac{\mathfrak{Q}(\ell q\, t)}{\varphi(t)}, \qquad t = |G_g(P)|,\ q = |\det(P, P')|/\ell. \tag{9}$$

No joint law of the two residues under the shared w is needed.

Step 5 (the partner's rational weight). Fix $P = (a, b)$ and P' with $|\Delta| = \ell q$. The algebraic factor in (9) depends on g only through $n := G_g(P)/\ell$, and $Y_P = 1$ iff $n \in \mathcal{N}_\ell$. For fixed n, the g with $G_g(P) = \ell n$ are $g^* + j(a, b)$ (Lemma 8.6), at most $2M/V$ values of j. Along the fibre, $G_g(P') = G_{g^*}(P') - j\Delta$, so $n' = G_g(P')/\ell$ runs through a progression of difference q in an interval of length at most $2qM/V$. The filter requires $|n'| \in [X_r/256, X_r]$. For each sign, enlarge the n'-interval to length $\max(\cdot, X_r B^{-C_0})$ inside $[X_r B^{-C_0}, X_r]$. Lemma 9.4 (modulus $q \le 2S^2 \le B^{C_0}$) gives

$$\#\{g \in \mathcal{G} : G_g(P) = \ell n,\ Y_{P'}(g) = 1\} \le U := \frac{3M}{V_-} \rho_r (\log N)^C,$$

uniformly in n, using $X_r B^{-C_0}/q \le M/V$. Hence

$$\sum_g Y_P Y_{P'} \frac{\mathfrak{Q}(\ell q t)}{\varphi(t)} \le U \sum_{n \in \mathcal{N}_\ell} \frac{\mathfrak{Q}(\ell^2 |n| q)}{\varphi(\ell |n|)}.$$

Step 6 (partners). For fixed P and q there are at most 12 points P' (Lemma 9.1).

Step 7 (assembly). Bound the ℓ-dependent geometric factors by their envelopes: $\#\mathcal{P}_\ell \le C V_{\max}^2$, $|\mathcal{G}| \ge cM^2$, $V^{-(2d-4)} \le V_-^{-(2d-4)}$. Then apply (8):

$$\mathbb{E}_{\ell, A}[N_\ell(N_\ell - 1)] \le \frac{C}{H^2 V_-^{2d-4}} \cdot \frac{V_{\max}^2}{M^2} \cdot \frac{M \rho_r}{V_-} (\log N)^C \sum_{q \le 2S^2} \tau(q) \frac{\Psi(X'_+) \Psi(X^*)^2}{Y q (MV_-)^2}.$$

Insert $\Psi(X'_+) = X'_+ \rho_r = (2MV_{\max}/Y)\rho_r$ and $\Psi(X^*) = X^* \rho_F \le CMHV_{\max}^d \rho_F$. The powers of M and H cancel exactly:

$$\frac{1}{H^2 V^{2d-4}} \cdot \frac{V^2}{M^2} \cdot \frac{M}{V} \cdot \frac{MV}{Y} \cdot \frac{M^2 H^2 V^{2d}}{Y (MV)^2} = \frac{V^4}{Y^2} \asymp S^4,$$

up to constants from $V_- \asymp V_{\max}$. Finally $\sum_{q \le 2S^2} \tau(q)/q \le (1 + \log 2S^2)^2 = (\log N)^{O(1)}$. □

Remark 9.8. The residue is integrated out over w (Step 4) *before* any arithmetic estimate is applied. After that, the algebraic factor depends on g only through n, and this is what lets Step 5 sum the partner's rational indicator along a fibre by divisor switching. In the design without w (i.e. $F(g_0, g_1) = N$), the forced residue moves with the fibre index together with the partner indicator (Proposition 3.16), and we do not know how to prove the corresponding bound.

9.6. **Relation supply.**

Theorem 9.9 (relation supply)**.** *For $N \geq N_0$, $\gamma = \Pr_{A,\ell}[N_\ell(A) \geq 1] \geq L^{-o(1)}$. More precisely, with $m := S^2\rho_r\rho_F = L^{o(1)}$ and a fixed polylogarithm $\mathcal{P} = (\log N)^C$, $\gamma \geq 1/(\mathcal{P}^3 + \mathcal{P}/m)$.*

Proof. By Cauchy–Schwarz (Paley–Zygmund) for the nonnegative integer variable $N_\ell(A)$ under the product law of (ℓ, A), $\gamma \geq (\mathbb{E}N)^2/\mathbb{E}N^2 = (\mathbb{E}N)^2/(\mathbb{E}N(N-1) + \mathbb{E}N)$. The function $z \mapsto z^2/(K+z)$ is increasing on $z > 0$. By Theorems 8.7 and 9.7, $\mathbb{E}N \geq m/\mathcal{P}$ and $\mathbb{E}N(N-1) \leq \mathcal{P}m^2$. So $\gamma \geq (m/\mathcal{P})^2/(\mathcal{P}m^2 + m/\mathcal{P})$. Finally $m = L^{o(1)}$ by (2). □

Remark 9.10. m is in general smaller than every negative power of $\log N$: the leading terms of $\log(1/\rho_r) + \log(1/\rho_F)$ and $\log S^2$ cancel, but the second-order terms of the de Bruijn expansion do not. So $\gamma \geq 1/(\log N)^{O(1)}$ need not hold for our explicit S. Only $\gamma \geq L^{-o(1)}$ is used.

9.7. **A rational pair correlation (not used in the proof).** For comparison we record the purely rational version of Step 5, which explains why the rational side is randomised in two dimensions. Put $R_1 := \mathbb{E}_{\ell,g} \sum_P Y_P$ and $R_2 := \mathbb{E}_{\ell,g} \sum_{P \neq P'} Y_P Y_{P'}$ (sums over $\mathcal{P}_\ell \cap \Lambda_\ell(g)$).

Proposition 9.11 ("Theorem R")**.** *$R_1 \geq S^2\rho_r/(\log N)^C$ and $R_2 \leq (\log N)^C S^4 \rho_r^2$.*

Proof. The lower bound is the rational part of Theorem 8.7. For the upper bound fix ℓ, P and P' with $|\Delta| = \ell q$. The map $g \mapsto (n, n') = (G_g(P)/\ell, G_g(P')/\ell)$ on $\{g : \ell \mid G_g(P)\}$ is injective, because the matrix $\left(\begin{smallmatrix} a & -b \\ a' & -b' \end{smallmatrix}\right)$ has determinant $-\Delta \neq 0$. Its image is the lattice $\Gamma_q = \{(n, n') : n' \equiv \lambda n \ (q)\}$, $\lambda \equiv b'b^{-1}$, of index q ($b > 2S^2 \geq q$ is prime, so b is a unit modulo q). Indeed $bG_g(P') - b'G_g(P) = -g_1\Delta$, so the image lies in Γ_q, and both lattices have index q. The image of $\mathcal{G}$ lies in a parallelogram Π of area $M^2q/(4\ell)$ inside $[-X_r, X_r]^2$, with one side of length $\asymp MV/\ell$ in a direction making an angle in $[\arctan\frac{1}{2}, \arctan 2]$ with the axes. Cover Π by $O(qS^2)$ squares of side $w_0 = M/V$. In a square $I_1 \times I_2$ away from the axes, Lemma 9.4 in each coordinate (modulus q, length $w_0 \geq X_r/(2S^2)$) gives at most $\sum_{r \bmod q}\big((w_0/q)\rho_r(\log N)^C\big)^2 = (w_0^2/q)\rho_r^2(\log N)^{2C}$ smooth points of Γ_q. Near the axes the filter $|n|, |n'| \geq X_r/256$ removes everything. So each pair has at most $(M^2/\ell)\rho_r^2(\log N)^C$ admissible g. There are at most $200S^2$ partners P' per P (they lie in the regular lattice $\Lambda_\ell((a,b)) \cap [-V, V]^2$), and $\#\mathcal{P}_\ell \leq CV^2$. Dividing by $|\mathcal{G}| \asymp M^2$ gives $R_2 \leq CV^2S^2\ell^{-1}\rho_r^2(\log N)^C \leq (\log N)^C S^4\rho_r^2$. □

In the one-dimensional design $G = x - my$ the rational values (s, s') of a pair move on a *line* $b's - bs' = \Delta/\ell$ as m varies. Then divisor switching reduces to counting solutions of one linear equation in a box, which has no area main term, and we know no bound of the needed strength. In two dimensions the pair fills a lattice region of area much larger than its perimeter times the lattice scale, and that is all the argument needs.

10. Post-processing at fixed degree

This section adapts the number field sieve post-processing of Buhler–Lenstra–Pomerance [6] and Lee–Venkatesan [29, §6] to our rows. Because d is fixed, every field-dependent quantity is $N^{O(1)}$. This simplifies matters considerably: there are $O(\log N)$ character columns, at most one exceptional square class, and the square root needs no lattice reduction.

10.1. **Field-level events.**

Lemma 10.1 (irreducibility certificate)**.** *Let $g \in \mathcal{G}$ with $\gcd(g_1, N) = 1$, let w be a prime in $(W, 2W]$, and let N have no prime factor $\leq B$. Let $\mathcal{P}_{\mathrm{IR}}$ be the set of primes in $(r_0(d), \frac{1}{2}\log H]$,*

where $r_0(d) \geq 16$ is a constant. For c uniform on C_H, the probability that no $r \in \mathcal{P}_{\mathrm{IR}}$ has $f \bmod r$ of degree d and irreducible over $\mathbb{F}_r$ is at most

$$e\,(1 - 1/(4d))^{|\mathcal{P}_{\mathrm{IR}}|} \;\leq\; \exp\bigl(-\log H/(13d \log\log H)\bigr) \qquad (N \geq N_0).$$

If some r certifies, then f is irreducible over $\mathbb{Q}$.

Proof. Let $Q_{\mathrm{IR}} = \prod_{r \in \mathcal{P}_{\mathrm{IR}}} r = H^{1/2+o(1)} \leq (2H+1)/d$. Every residue class of C_H modulo Q_{IR} has between $\lfloor (2H+1)/Q_{\mathrm{IR}} \rfloor^d$ and $\lceil (2H+1)/Q_{\mathrm{IR}} \rceil^d$ elements, so the law of $c \bmod Q_{\mathrm{IR}}$ is at most $(1+1/d)^d \leq e$ times uniform. Under the uniform law the reductions modulo distinct r are independent (CRT). Fix $r \in \mathcal{P}_{\mathrm{IR}}$; note $r < B$, so $r \nmid Nw$. If $r \nmid g_1$, then $c \mapsto f \bmod r$ is an affine bijection from $\mathbb{F}_r^d$ onto the polynomials of degree at most d with $\phi(m_r) = Nwg_1^{-d} \neq 0$, where $m_r = g_0 g_1^{-1} \bmod r$: multiplication by $g_1 x - g_0$ maps polynomials of degree at most $d-1$ bijectively onto those vanishing at m_r. If $r \mid g_1$, then $g_1 x - g_0 \equiv -g_0$ is a unit, and $f \bmod r$ ranges over all polynomials with leading coefficient $a_d \not\equiv 0$ (as $a_d g_0^d \equiv Nw$). Monic irreducible polynomials of degree d over $\mathbb{F}_r$ with prescribed constant term $v \neq 0$ correspond to the elements of degree d of $\mathbb{F}_{r^d}$ with norm $(-1)^d v$. There are at least $((r^d-1)/(r-1) - 2r^{d/2})/d \geq r^{d-1}/(2d)$ of them. After the translation $x \mapsto x + m_r$ and summing over the $r-1$ leading coefficients, the irreducible fraction is at least $(r-1)/(2dr) \geq 1/(4d)$ in the first case, and at least $1/(2d)$ in the second. Hence the bound, with $|\mathcal{P}_{\mathrm{IR}}| \geq \log H/(3 \log\log H)$. A factorisation of the primitive part of f over $\mathbb{Z}$ would reduce modulo r to a factorisation with positive degrees, since $r \nmid f_d$. □

Lemma 10.2 (rejection events)**.** *Under the raw law $\mathcal{A}$, let $\mathcal{R}$ be the union of the events: (i) no irreducibility certificate; (ii) $N \mid f'(m)$; (iii) the content of f equals w. Then* $\Pr(\mathcal{R}) \leq \exp(-\log H/(13d \log\log H)) + 2/(2H+1) + 1/W = L^{-\omega(1)}$*. In particular* $\Pr(\mathcal{R}) \leq \gamma/2$ *for large N, by Theorem 9.9.*

Proof. (i) is Lemma 10.1. (ii) Modulo N, $f'(m) \equiv \hat{f}'(m) + g_1 R_c(m,1)$, because the derivative of $(g_1 x - g_0) R_c(x,1)$ is $g_1 R_c(x,1) + (g_1 x - g_0)\partial_x R_c(x,1)$ and $g_1 m - g_0 \equiv 0$. Here $R_c(m,1) = \sum_i c_i m^{d-1-i}$ has coefficient 1 at c_{d-1} and g_1 is a unit. So for fixed $c_0, \ldots, c_{d-2}$ at most one $c_{d-1} \in [-H, H]$ gives $N \mid f'(m)$, since $2H+1 < N$. (iii) If the content is w, then $w \mid f_d = a_d + g_1 c_0$, and g_1 is invertible modulo the prime $w > M$. For fixed $(g, w, c_1, \ldots, c_{d-1})$ this fixes c_0 modulo w, so the probability is at most $\lceil (2H+1)/w \rceil/(2H+1)$. Finally $\log H \asymp \log N$, so the bound in (i) is $\exp(-\Omega(\log N/\log\log N)) = L^{-\omega(1)}$. □

Remark 10.3. Rejecting a field with content w loses nothing: then every nonzero value $F_A(P)$ is divisible by the prime $w > B$, so $N_\ell(A) = 0$ for all ℓ. Also, the leading coefficient never causes a degree drop ($f_d \not\equiv 0 \bmod g_1$), and a non-unit leading coefficient is a factor exit because $0 < |f_d| < N$. So no restriction to unit leading coefficients is needed.

10.2. **Rows and the row identity.** Fix an accepted field: $\gcd(g_1 f_d w, N) = 1$, f is irreducible and primitive, and $f_d^{d-2} f'(m)$ is a unit modulo N. Let α be a root of f, $K = \mathbb{Q}(\alpha)$, $\beta := f_d \alpha$, and let $g(y) := f_d^{d-1} f(y/f_d) \in \mathbb{Z}[y]$ be the monic minimal polynomial of β. Put $D := f_d \operatorname{disc}(f)$. Since $\operatorname{disc}(g) = f_d^{(d-1)(d-2)} \operatorname{disc}(f)$, the primes dividing $\operatorname{disc}(g)$ divide D. Define the ring homomorphism

$$\psi : \mathbb{Z}[\beta] \to \mathbb{Z}/N, \qquad \beta \mapsto f_d m, \qquad m = g_0 g_1^{-1} \bmod N.$$

It is well defined because $g(f_d m) = f_d^{d-1} f(m) \equiv f_d^{d-1} g_1^{-d} N w \equiv 0$. For an atom $P = (a, b)$ put $z_P := f_d a - b\beta = f_d(a - b\alpha) \in \mathbb{Z}[\beta]$. Then

$$\mathrm{N}_{K/\mathbb{Q}}(z_P) = f_d^{d-1} F_A(P), \qquad \psi(z_P) = f_d G(P)/g_1.$$

Write $G(P) = \sigma_P \ell n_P$. Then $\ell \equiv \sigma_P g_1 f_d^{-1} \psi(z_P)/n_P \pmod N$. A row i comes from $R_i = r_i^2$ with primitive split $R_i \equiv u_0/v_0$, signs $e_{i\ell} = \pm 1$ and atoms P_ℓ, so

$$r_i^2 \equiv \mathfrak{q}_i \, f_d^{-E_i} \prod_\ell \psi(z_{P_\ell})^{e_{i\ell}} \pmod N,$$
$$\mathfrak{q}_i := \operatorname{sgn}(u_0 v_0) \prod_\ell \big(\sigma_{P_\ell} g_1 / n_{P_\ell}\big)^{e_{i\ell}} \in \mathbb{Q}^\times, \qquad E_i := \sum_\ell e_{i\ell}. \tag{10}$$

The factor $\operatorname{sgn}(u_0 v_0)$ is necessary: for example $R = -1$ can have shortest vector $(1, -1)$, with an empty product of atoms. The same atom can occur in several rows. For a set S of rows put $e_P(\mathsf{S}) := \sum_{i \in \mathsf{S}} e_{iP} \in \mathbb{Z}$ and $E(\mathsf{S}) := \sum_P e_P(\mathsf{S})$.

10.3. **Columns.** The matrix has one row per row of the algorithm and the following columns, all over $\mathbb{F}_2$:

(a) the sign of $\mathfrak{q}_i$;
(b) for each prime $p \le B$: $\sum_\ell e_{i\ell} v_p(n_{P_\ell})$;
(c) one parity column E_i;
(d) for each prime $r \le B$ with $r \nmid D$ and each root s of $f \bmod r$: $\sum_\ell e_{i\ell} \, [a_{P_\ell} \equiv b_{P_\ell} s \ (r)] \, v_r(F_A(P_\ell))$;
(e) t_χ character columns (Section 10.5).

The prime factors of g_1 are never needed: if (a)–(c) vanish on S, then $E(\mathsf{S})$ is even and $\mathfrak{q}_\mathsf{S} := \prod_{i \in \mathsf{S}} \mathfrak{q}_i = g_1^{E(\mathsf{S})} \prod_P n_P^{-e_P(\mathsf{S})}$ is a positive rational square, with the explicit square root $g_1^{E(\mathsf{S})/2} \prod_{p \le B} p^{-\frac{1}{2}\sum_P e_P(\mathsf{S}) v_p(n_P)}$. This parity column also makes $f_d^{-E(\mathsf{S})}$ a square. It is the device of [6, §12.7].

Lemma 10.4 (good primes)**.** *Let r be a prime with $r \nmid D$, and let (a, b) be coprime with $F_A(a, b) \neq 0$.*

(i) If $r \mid b$, then $r \nmid F_A(a, b)$, and $a - b\alpha$ is a unit at every prime of $\mathcal{O}_K$ above r.
(ii) If $r \nmid b$, the only prime $\mathfrak{p} \mid r$ with $\operatorname{ord}_\mathfrak{p}(a - b\alpha) \neq 0$ is the degree-one prime corresponding to the root $s \equiv ab^{-1}$ of $f \bmod r$; it occurs only if $r \mid F_A(a, b)$, and then $\operatorname{ord}_\mathfrak{p}(a - b\alpha) = v_r(F_A(a, b))$.

Consequently, if columns (d) vanish on S, then $\gamma_\mathsf{S} := \prod_P (a_P - b_P \alpha)^{e_P(\mathsf{S})}$ has even valuation at every prime of $\mathcal{O}_K$ not dividing D.

Proof. As $r \nmid f_d$, α is r-integral, and $\mathbb{Z}_{(r)}[\alpha]$ has discriminant $\operatorname{disc}(f) f_d^{-(2d-2)}$, an r-unit. So $\mathbb{Z}_{(r)}[\alpha]$ is the maximal order at r, and by Dedekind–Kummer the primes above r correspond to the irreducible factors ϕ_j of $f \bmod r$ and are unramified. (i) $F_A(a, b) \equiv f_d a^d \not\equiv 0$, and $a - b\alpha \equiv a$ modulo every $\mathfrak{p} \mid r$. (ii) $a - b\alpha \in \mathfrak{p}_j$ iff $\phi_j(ab^{-1}) \equiv 0$, i.e. iff $\phi_j = x - ab^{-1}$. Since $\mathrm{N}(a - b\alpha) = F_A(a, b)/f_d$ and $\mathfrak{p}$ has degree one, $\operatorname{ord}_\mathfrak{p}(a - b\alpha) = v_r(F_A(a, b))$. For the consequence: primes $r > B$ with $r \nmid D$ contribute nothing, since $F_A(P)$ is B-smooth. □

10.4. **The residual square-class group.** Let S_D be the set of primes of $\mathcal{O}_K$ above primes dividing D. Put $V_D := \{\gamma \in K^\times : \operatorname{ord}_\mathfrak{P} \gamma \text{ even for all } \mathfrak{P} \notin S_D\}$ and $\mathsf{H} := V_D / K^{\times 2}$. By Lemma 10.4, every S in the kernel of columns (a)–(d) has $\gamma_\mathsf{S} \in V_D$, and $\phi(\mathsf{S}) := [\gamma_\mathsf{S}] \in \mathsf{H}$ is $\mathbb{F}_2$-linear in S, because $\gamma_{\mathsf{S}+\mathsf{S}'} = \gamma_\mathsf{S} \gamma_{\mathsf{S}'} / \gamma^2_{\mathsf{S} \cap \mathsf{S}'}$.

Lemma 10.5 (size of H; cf. [6, Thm. 6.7])**.** *$|\mathsf{H}| \le 2^{d\omega(D)+d} h_K$, and $\log |\mathsf{H}| = O_d(\log N)$.*

Proof. Let $W_0 := \{\gamma : \gamma \mathcal{O}_K = \mathfrak{a}^2\}$ and $Y_0 := \mathcal{O}_K^\times K^{\times 2}$. Then $V_D \supset W_0 \supset Y_0 \supset K^{\times 2}$. The map $V_D \to \bigoplus_{\mathfrak{P} \in S_D} \mathbb{Z}/2$, $\gamma \mapsto (\operatorname{ord}_\mathfrak{P} \gamma \bmod 2)$, has kernel W_0. The map $W_0 \to \mathrm{Cl}(K)[2]$, $\gamma \mapsto [\mathfrak{a}]$, is well defined (the ideal group is torsion-free) with kernel Y_0. And $Y_0 / K^{\times 2} \cong \mathcal{O}_K^\times / \mathcal{O}_K^{\times 2}$ has

order $2^{r_1+r_2} \le 2^d$, since the only roots of unity are ± 1 (d odd). Also $|S_D| \le d\,\omega(D)$. For the size: $\|f\|_\infty \le M(H+1) < N$ and d is fixed, so $\log|D|$ and $\log|\operatorname{disc} g|$ are $O_d(\log N)$. Every ideal class contains an integral ideal of norm at most $\sqrt{|\Delta_K|} \le \sqrt{|\operatorname{disc} g|}$, and there are at most $X(1+\log X)^{d-1}$ integral ideals of norm at most X. So $\log h_K = O_d(\log N)$. □

10.5. **Characters.** For $0 \ne \mathfrak{q} \in \mathsf{H}$ let $L_\mathfrak{q} := K(\sqrt{\mathfrak{q}})$, of degree $2d$. Put $\bar\Delta := |\operatorname{disc} g|^2\, 8^d |D|^d$.

Lemma 10.6. *$|\Delta_{L_\mathfrak{q}}| \le \bar\Delta$, and the quadratic subfield F of $L_\mathfrak{q}$, if any, has $|\Delta_F| \le \bar\Delta^{1/d}$.*

Proof. $\Delta_L = \Delta_K^2\, \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{d}_{L/K})$. A representative $\gamma \in V_D$ has even valuation at every $\mathfrak{P} \nmid 2D$, so L/K is unramified there. At $\mathfrak{P} \mid D$ with $\mathfrak{P} \nmid 2$, $\operatorname{ord}_\mathfrak{P} \mathfrak{d} \le 1$, and at $\mathfrak{P} \mid 2$, $\operatorname{ord}_\mathfrak{P} \mathfrak{d} \le 2\operatorname{ord}_\mathfrak{P}(2)+1$. So $\mathrm{N}(\mathfrak{d}) \le 8^d|D|^d$, and $|\Delta_K| \le |\operatorname{disc} g|$. For $F \subset L$, $|\Delta_F|^{[L:F]}$ divides Δ_L, and $[L:F] = d$. □

Fact 10.7 (Stark [50, Lemma 8]; [29, Fact 6.22])**.** *Let L be a number field of degree n, and $c(L) = 4$ if $L/\mathbb{Q}$ is normal, $c(L) = 4\cdot n!$ otherwise. If ζ_L has a real zero β_0 with $1 - (c(L)\log|\Delta_L|)^{-1} \le \beta_0 < 1$, then $\zeta_F(\beta_0) = 0$ for some quadratic field $F \subset L$.*

Fact 10.8 (Landau; see [11, Ch. 14], [29, Fact 6.24])**.** *There is an effective absolute constant $c_L > 0$ such that for distinct real primitive characters χ_1, χ_2 of conductors q_1, q_2, $L(s,\chi_1)L(s,\chi_2)$ has at most one real zero (with multiplicity) in $(1 - c_L/\log(q_1q_2), 1)$.*

Fact 10.9 (Lagarias–Odlyzko [28, Thm. 1.3])**.** *Let L/K be Galois with group G, $n_L = [L:\mathbb{Q}]$, $d_L = |\Delta_L|$, and $\mathcal{C}$ a conjugacy class. If $x \ge \exp(10 n_L (\log d_L)^2)$, then*

$$\Big|\pi_\mathcal{C}(x) - \frac{|\mathcal{C}|}{|G|}\mathrm{Li}(x)\Big| \le \frac{|\mathcal{C}|}{|G|}\mathrm{Li}(x^{\beta_0}) + c_1 x \exp\big(-c_2 n_L^{-1/2}(\log x)^{1/2}\big),$$

with absolute effective c_1, c_2. The β_0 term is present only if ζ_L has a real zero $\beta_0 \in [1 - (4\log d_L)^{-1}, 1)$. Here $\pi_\mathcal{C}(x)$ counts primes $\mathfrak{p}$ of K unramified in L with $\mathrm{N}\,\mathfrak{p} \le x$ and Frobenius in $\mathcal{C}$.

Put $\nu_1 := \min\big(1/(4(2d)!\log\bar\Delta),\ c_L/(2\log\bar\Delta)\big)$, and let $\mathsf{E} := \{0 \ne \mathfrak{q} \in \mathsf{H} : \zeta_{L_\mathfrak{q}}$ has a real zero in $[1-\nu_1, 1)\}$.

Lemma 10.10 (at most one exceptional class)**.** *$|\mathsf{E}| \le 1$ for every field.*

Proof. Let $\mathfrak{q} \in \mathsf{E}$ with zero β_0. By Fact 10.7 (with $c = 4(2d)!$, valid in all cases) there is a quadratic $F_\mathfrak{q} \subset L_\mathfrak{q}$ with $\zeta_{F_\mathfrak{q}}(\beta_0) = 0$, and as $\zeta(\beta_0) \ne 0$ on $(0,1)$, $L(\beta_0, \chi_{F_\mathfrak{q}}) = 0$. Since $[L_\mathfrak{q}:\mathbb{Q}] = 2d$ with d odd, $L_\mathfrak{q}$ has at most one quadratic subfield (two would generate a biquadratic subfield, and $4 \nmid 2d$), and K has none. Hence $L_\mathfrak{q} = KF_\mathfrak{q}$. The map $\mathfrak{q} \mapsto L_\mathfrak{q}$ is injective by Kummer theory, so $\mathfrak{q} \mapsto F_\mathfrak{q}$ is injective. If $\mathfrak{q} \ne \mathfrak{q}'$ were both in E, then $\chi_{F_\mathfrak{q}}, \chi_{F_{\mathfrak{q}'}}$ would be distinct real primitive characters with conductors at most $\bar\Delta^{1/d}$ (Lemma 10.6), each with a real zero $\ge 1 - \nu_1 > 1 - c_L/\log(\bar\Delta^{2/d})$. That contradicts Fact 10.8. □

Sampler. Let x be the least integer with $\log(x/2) \ge \max(20d(\log\bar\Delta)^2, (\log N)^2)$, so $\log x = O_d((\log N)^2)$. Draw r uniform in $(x/2, x]$. If r is prime (AKS), find the set Z_r of roots of f mod r (Las Vegas, e.g. Cantor–Zassenhaus [8]). With probability $|Z_r|/d$ output (r, s) with s uniform in Z_r; otherwise redraw. Each pair (r,s) is output with the same probability, so the output is uniform over the degree-one primes $\mathfrak{p} = (r, \alpha - s)$ with $\mathrm{N}\,\mathfrak{p} \in (x/2, x]$. Since $r > \bar\Delta \ge |D|$, $r > B$, and r exceeds every $|F_A(P)|$, no atom $a - b\alpha$ lies in $\mathfrak{p}$.

The expected number of draws is $O(d\log x)$. To see this, apply Fact 10.9 to K/K (trivial group, $n_L = d$). If ζ_K had a real zero in $[1 - (4\cdot d!\log|\Delta_K|)^{-1}, 1)$, Fact 10.7 would produce a quadratic subfield of K, which does not exist. A zero in the wider interval of Fact 10.9 contributes at most $x^{1-1/(4d!\log\bar\Delta)} = o(x/\log x)$ because $\log x \ge (\log N)^2$. So there are at least $x/(3\log x)$ prime ideals of K with norm in $(x/2, x]$; those of degree ≥ 2 number at most $d\sqrt{x}$.

Characters. A degree-one prime $\mathfrak{p} = (r, \alpha - s)$ with $r \nmid D$ defines a quadratic character on H: $\chi_{\mathfrak{p}}(\mathfrak{q}) :=$ the Legendre symbol modulo $\mathfrak{p}$ of a $\mathfrak{p}$-unit in the class $\gamma\mathfrak{p}^{-\operatorname{ord}_{\mathfrak{p}}\gamma}$, which is well defined on $V_D/K^{\times 2}$. On atoms, $\chi_{\mathfrak{p}}(a - b\alpha) = \big((a - bs)/r\big)$. The character columns are $\sum_{\ell} e_{i\ell}(1 - \chi_{\mathfrak{p}_j}(a_{P_\ell} - b_{P_\ell}\alpha))/2 \bmod 2$, for independent samples $\mathfrak{p}_1, \dots, \mathfrak{p}_{t_\chi}$.

Lemma 10.11 (characters separate)**.** *For every field and every $\mathfrak{q} \in \mathsf{H} \setminus (\mathsf{E} \cup \{0\})$, $\Pr_{\mathfrak{p}}[\chi_{\mathfrak{p}}(\mathfrak{q}) = -1] \geq 1/(5d)$ for $N \geq N_0(d)$. With $t_\chi := \lceil 5d(\ln|\mathsf{H}|_{\mathrm{bd}} + 2\ln\log N)\rceil = O_d(\log N)$, where $|\mathsf{H}|_{\mathrm{bd}}$ is the bound of Lemma 10.5, the probability that $\mathsf{H}_0 := \mathsf{H} \cap \bigcap_j \ker\chi_{\mathfrak{p}_j}$ is not contained in $\mathsf{E} \cup \{0\}$ is at most $(\log N)^{-2}$. Otherwise $|\mathsf{H}_0| \leq 2$.*

Proof. Apply Fact 10.9 to $L_{\mathfrak{q}}/K$ with $G = C_2$, $\mathcal{C} = \{\sigma\}$, $n_L = 2d$, $d_L \leq \bar{\Delta}$. For $\mathfrak{p} \nmid 2D$, $\chi_{\mathfrak{p}}(\mathfrak{q}) = -1$ iff the Frobenius of $\mathfrak{p}$ is σ. The main term contributes $\frac{1}{2}(\mathrm{Li}(x) - \mathrm{Li}(x/2)) \geq x/(4\log x)(1 - o(1))$ inert primes of norm in $(x/2, x]$. A real zero, if present, is $< 1 - \nu_1$ because $\mathfrak{q} \notin \mathsf{E}$, and contributes at most $2x^{1-\nu_1} \leq 2x\exp(-\nu_1(\log N)^2) = o(x/\log x)$, since $\nu_1 \gg_d 1/\log N$. The error term is $2c_1 x\exp(-c_2\log N/\sqrt{2d}) = o(x/\log x)$, as $\log x = O((\log N)^2)$. Primes of degree ≥ 2 and ramified primes are negligible. There are at most $d(\pi(x) - \pi(x/2)) \leq 0.6dx/\log x$ degree-one primes in all, so the ratio is at least $1/(5d)$. Independence and a union bound over H give the probability. Finally H_0 is a subgroup contained in $\mathsf{E} \cup \{0\}$, so $|\mathsf{H}_0| \leq 2$. □

10.6. **Square roots.**

Algorithm 10.12 (SQ(Γ); after [6, §9])**.** Input: $\Gamma = g'(\beta)^2 \prod_P z_P^{|e_P|} \in \mathbb{Z}[\beta]$, and the bound A_{sq} of Lemma 10.13.

1. Let p be the least prime in $(B, 2B]$ with $p \nmid D$, found by sieving $(B, 2B]$ once per field. Then $g \bmod p$ is squarefree, and Γ is a unit at p, because $\mathrm{N}(\Gamma) = \pm\operatorname{disc}(g)^2 \prod (f_d^{d-1} F_A(P))^{|e_P|}$ has only prime factors $\leq B$ or dividing D.
2. Factor $g \bmod p = \phi_1 \cdots \phi_\tau$ over $\mathbb{F}_p$. In each $\mathbb{F}_p[y]/(\phi_j)$ test whether Γ is a square and compute a square root ρ_j (Tonelli–Shanks). If some component is a non-square, output NONSQUARE.
3. For each of the $2^{\tau-1}$ sign vectors ϵ with $\epsilon_1 = +1$: combine the $\epsilon_j\rho_j$ by the Chinese remainder theorem to $\rho_\epsilon \in \mathbb{Z}[\beta]/p$; Newton-lift the inverse square root from $z_0 = \rho_\epsilon^{-1}$ by $z \mapsto z + z(1 - \Gamma z^2)/2$ to precision $p^e > 2A_{\mathrm{sq}}$; put $\eta_\epsilon := \Gamma z \bmod p^e$ with centred coefficients. If $\eta_\epsilon^2 = \Gamma$ exactly, output η_ϵ.
4. Output NONSQUARE.

Lemma 10.13 (square roots)**.** *Let* $\mathrm{Mh} := \lfloor\|g\|_2\rfloor + 1$, $g_{\mathrm{d}} := d\|g\|_1\mathrm{Mh}^{d-1}$, $U := g_{\mathrm{d}}^2 \prod_P (\|z_P\|_1\mathrm{Mh}^{d-1})^{|e_P|}$ *and* $A_{\mathrm{sq}} := d\lceil U^{1/2}\rceil 2^{d-1}\mathrm{Mh}\, g_{\mathrm{d}}^{d-1}$. *Then:*

(i) Any output η satisfies $\eta^2 = \Gamma$. If $\Gamma \in K^{\times 2}$, then $\Gamma = \eta^2$ with $\eta \in \mathbb{Z}[\beta]$, $\|\eta\|_\infty \leq A_{\mathrm{sq}}$, and SQ outputs $\pm\eta$.

(ii) $\log A_{\mathrm{sq}} = O_d((1 + \sum_P |e_P|)\log N)$, and SQ runs in $O_d(\mathsf{M}(\log A_{\mathrm{sq}})\log\log A_{\mathrm{sq}}) + (\log N)^{O(1)}$ bit operations (Las Vegas), where M is the cost of integer multiplication.

Proof. (i) If $\Gamma = \delta^2$ with $\delta \in K$, then $(\delta/g'(\beta))^2 = \prod z_P^{|e_P|} \in \mathcal{O}_K$, so $\delta/g'(\beta) \in \mathcal{O}_K$. Moreover $g'(\beta)\mathcal{O}_K \subset \mathbb{Z}[\beta]$, because $\mathcal{O}_K \subset \mathcal{O}_K^\vee \subset \mathbb{Z}[\beta]^\vee = g'(\beta)^{-1}\mathbb{Z}[\beta]$ (Euler's lemma). So $\eta := \delta \in \mathbb{Z}[\beta]$. Since $p \nmid \operatorname{disc} g$, $\mathbb{Z}_p[\beta] \cong \prod_j \mathbb{Z}_p[y]/(G_j)$ with unramified local factors lifting the ϕ_j. The reduction of η is $\pm\rho_\epsilon$ for exactly one ϵ with $\epsilon_1 = +1$. A unit square root with prescribed residue lifts uniquely (p odd), so $\eta_\epsilon \equiv \pm\eta \pmod{p^e}$, and centring recovers $\pm\eta$ since $p^e > 2A_{\mathrm{sq}}$. NONSQUARE at step 2 is correct, since η would reduce to a square in each residue field $\mathcal{O}_K/\mathfrak{P} = \mathbb{F}_p[y]/(\phi_j)$ (p is prime to the index). The coefficient bound: the conjugates satisfy $|\sigma_k(z)| \leq \|z\|_1\mathrm{Mh}^{d-1}$ and $|g'(\beta_k)| \leq g_{\mathrm{d}}$, so $|\sigma_k(\eta)| \leq U^{1/2}$. The coefficients of η are recovered

with the inverse Vandermonde matrix, $(V^{-1})_{jk} = [y^j](g(y)/(y-\beta_k))/g'(\beta_k)$. Its numerators are at most $2^{d-1}\mathrm{Mh}$ in absolute value, and $\prod_k |g'(\beta_k)| = |\operatorname{disc} g| \geq 1$ gives $|g'(\beta_k)| \geq g_{\mathrm{d}}^{-(d-1)}$. (ii) The bound on $\log A_{\mathrm{sq}}$ is immediate. The cost consists of a product tree for Γ, Newton iteration doubling the precision with $O(d^2)$ multiplications per step, and $2^{d-1} = O(1)$ sign vectors. □

10.7. **Linear algebra and the congruence.**

Algorithm 10.14 (POSTPROCESS(A, rows))**.**

1. Sample $\mathfrak{p}_1, \dots, \mathfrak{p}_{t_\chi}$ and build the $M_{\mathrm{rows}} \times C_{\mathrm{col}}$ matrix over $\mathbb{F}_2$ of columns (a)–(e).
2. Sample two uniform left-kernel vectors s_1, s_2 (Wiedemann's method over $\mathbb{F}_{2^e}$ with the Kaltofen–Saunders preconditioners [55, 26]). Verify them, and retry until they are linearly independent.
3. For $\mathsf{S} = s_1, s_2, s_1 + s_2$ in turn, run SQ on $\Gamma_{\mathsf{S}} := g'(\beta)^2 \prod_P z_P^{|e_P(\mathsf{S})|}$, and stop at the first S for which it returns η. If none does, go back to step 1 with fresh characters.
4. Return S and

$$\mathsf{Y}_{\mathsf{S}} := \sqrt{\mathfrak{q}_{\mathsf{S}}} \cdot f_d^{-E(\mathsf{S})/2} \cdot \psi(\eta) \cdot \big(\psi(\gamma_{\mathsf{S}}^-)\psi(g'(\beta))\big)^{-1} \bmod N, \qquad \gamma_{\mathsf{S}}^- := \prod_{e_P(\mathsf{S})<0} z_P^{|e_P(\mathsf{S})|}.$$

Proposition 10.15 (post-processing)**.** *Let $N \geq N_0$ and fix an accepted field and $M_{\mathrm{rows}} \geq C_{\mathrm{col}} + 12$ rows. Algorithm 10.14 is Las Vegas. Its expected time is $M_{\mathrm{rows}}^2 (\log N)^{O_d(1)} + B^{1+o(1)}$, and it returns a nonempty set S of rows and $\mathsf{Y}_{\mathsf{S}} \in (\mathbb{Z}/N)^\times$ with $\prod_{i\in\mathsf{S}} r_i^2 \equiv \mathsf{Y}_{\mathsf{S}}^2 \pmod N$. Its output and all its intermediate quantities are functions of the field, the certificates $(R_i, u_0, v_0, \mathit{atoms})$ and its own coins.*

Proof. Kernel sampling. The matrix has at most $M_{\mathrm{rows}}(2k \log_2 N + t_\chi + 2)$ nonzero entries. Wiedemann's method with the Kaltofen–Saunders preconditioners, run over $\mathbb{F}_{2^e}$ with $2^e \geq M_{\mathrm{rows}}^4$, returns a uniformly random kernel element of a black-box singular system in expected $O(M_{\mathrm{rows}} \cdot \#\mathrm{entries} + M_{\mathrm{rows}}^2)$ field operations [26, Thms. 2–4]. Here is how to sample uniformly from the binary kernel. Draw x uniform in $\mathbb{F}_2^{M_{\mathrm{rows}}}$, solve $A^\top y = A^\top x$ (padding $A^\top$ to a square matrix), and return the coordinate of $x - y$ at the basis element 1 of a fixed $\mathbb{F}_2$-basis of $\mathbb{F}_{2^e}$. Given $A^\top x$, the vector x is uniform on a coset of the kernel, and the solver sees only $A^\top x$; so the output is uniform on the binary kernel. Every output is verified. The kernel has dimension $r \geq 12$, so two samples are independent with probability $(1 - 2^{-r})(1 - 2^{1-r}) \geq 1 - 3 \cdot 2^{-12}$.

One of three is a square. Columns (a)–(d) vanish on s_1, s_2, so $\phi(s_1), \phi(s_2) \in \mathsf{H}$. Columns (e) vanish too, so they lie in H_0. If $|\mathsf{H}_0| \leq 2$ (probability $\geq 1 - (\log N)^{-2}$ by Lemma 10.11), either one of $\phi(s_1), \phi(s_2)$ is 0 or both equal the nonzero element, and then $\phi(s_1 + s_2) = 0$. In every case some $\mathsf{S} \in \{s_1, s_2, s_1 + s_2\}$ is nonempty with $\gamma_{\mathsf{S}} \in K^{\times 2}$. Then $\gamma'_{\mathsf{S}} := \prod_P z_P^{e_P(\mathsf{S})} = f_d^{E(\mathsf{S})} \gamma_{\mathsf{S}}$ is a square ($E(\mathsf{S})$ even), hence so is $\Gamma_{\mathsf{S}} = g'(\beta)^2 \gamma'_{\mathsf{S}} (\gamma_{\mathsf{S}}^-)^2$, and SQ returns η by Lemma 10.13. A failed round has probability at most $(\log N)^{-2} + 3 \cdot 2^{-12}$, so the expected number of rounds is $O(1)$.

The congruence. ψ is a ring homomorphism, so $\psi(\eta)^2 = \psi(g'(\beta))^2 \prod_P \psi(z_P)^{|e_P|}$. Hence

$$\mathsf{Y}_{\mathsf{S}}^2 = \mathfrak{q}_{\mathsf{S}} f_d^{-E(\mathsf{S})} \prod_P \psi(z_P)^{|e_P|} / \psi(\gamma_{\mathsf{S}}^-)^2 = \mathfrak{q}_{\mathsf{S}} f_d^{-E(\mathsf{S})} \prod_P \psi(z_P)^{e_P(\mathsf{S})} \equiv \prod_{i\in\mathsf{S}} r_i^2$$

by (10). Every quantity inverted is a unit modulo N. Indeed $\psi(z_P) = f_d G(P)/g_1$ with $G(P) = \pm \ell n_P$ coprime to N; $\psi(g'(\beta)) = f_d^{d-2} f'(m)$ by the field check; and $\sqrt{\mathfrak{q}_{\mathsf{S}}}$ involves only g_1 and primes $\leq B$.

Cost. Building the matrix and the characters costs $M_{\mathrm{rows}}(\log N)^{O(1)}$, and kernel sampling $M_{\mathrm{rows}}^2 (\log N)^{O(1)}$. Each SQ call costs $O(\mathsf{M}(kM_{\mathrm{rows}} \log N) \log\log N)$ by Lemma 10.13. The

Hensel prime costs $\tilde{O}(B)$, and $\sqrt{\mathfrak{qs}}$ costs $M_{\mathrm{rows}}(\log N)^{O(1)}$. No square root modulo N is ever extracted. The last claim of the proposition holds by construction: no step reads a root r_i. □

Remark 10.16 (relation to the literature)**.** The obstruction-group chain is [6, Thm. 6.7], with the index $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$ replaced by $2^{|S_D|}$. The random characters were suggested in [6, §8] and carried out rigorously in [29, §6]. The p-adic square root with exact verification is [6, §9], after [54]. Lee–Venkatesan's degree is $\asymp (\log N)^{1/3}$, which leads to $O(\log \log N)$ exceptional classes and an exceptional set of fields. At fixed degree, Lemma 10.10 gives at most one exceptional class for every field, and three tests suffice. We know of no explicit statement of this specialisation. Lenstra–Pomerance [33, §§3–4] use a similar explicit Hecke-character estimate with at most one exceptional zero, and random primes up to $\exp(c(\log|\Delta|)^2)$ to avoid GRH.

11. Batch smoothness

Lemma 11.1 (smooth parts; Bernstein [4, 5])**.** *Let $\mathcal{P}_B := \prod_{p \le B} p$, and let $x \ge 2$ and $2^e \ge \log_2 x$. The B-smooth part of x is $\gcd\big(x,\ (\mathcal{P}_B \bmod x)^{2^e} \bmod x\big)$. For given $x_1, \dots, x_n$, all their smooth parts can be computed in $O(\mathsf{M}(b) \log b)$ bit operations, where $b = \log_2 \mathcal{P}_B + \sum_i \log_2 x_i$.*

Proof. Write $x = s\,u$ with s smooth and $\gcd(u, \mathcal{P}_B) = 1$. Every $p^j \parallel s$ has $j \le \log_2 x \le 2^e$, so $s \mid \mathcal{P}_B^{2^e}$, while $\gcd(u, \mathcal{P}_B^{2^e}) = 1$. For the cost, form a product tree of the x_i, reduce $\mathcal{P}_B$ modulo the product, and descend a remainder tree; each level costs $O(\mathsf{M}(b))$ [53, §§10.1–10.3]. Then perform $e = O(\log \log x)$ modular squarings and a gcd for each x_i. □

Proposition 11.2 (the special-q search recognises $N_\ell(A)$ exactly)**.** SQ_SEARCH(A, ℓ) *is a deterministic function of (A, ℓ) and runs in time $\tilde{O}(B + S^2 \log N)$. If $\ell \nmid N$, then either it makes a factor exit, or it returns an atom if and only if $N_\ell(A) \ge 1$, and every atom it returns is a counted point.*

Proof. If $N_\ell(A) \ge 1$, the lattice is regular (Lemma 3.7), so step 1 does not fail. Step 2 enumerates at most $200S^2$ points from the reduced basis. It keeps exactly the points of $\mathcal{P}_\ell \cap \Lambda_\ell(g) \cap [-V, V]^2$ with $G(P) \ne 0$, $F_A(P) \ne 0$ and $|G(P)| \ge MV/32$. Every counted point is kept, and its gcd checks pass (its values are coprime to N). Step 3 computes all smooth parts exactly (Lemma 11.1). So step 4 returns a counted point whenever one exists, and only counted points. Lemma 11.1 applies with $b = O(B + S^2 \log N)$, since every value has $O(\log N)$ bits (Lemma 3.5). Trial division of the two chosen values costs $\tilde{O}(B)$. □

Remark 11.3. Since $B \le S^2$ when $\theta \le 1$, the batch costs $\tilde{O}(S^2)$ per box, the same as the enumeration. No smoothness test is probabilistic, and in particular no hyperelliptic curve method is used. (The published proof of the hyperelliptic curve method's running time [32] refers to an unpublished third part; Lee–Venkatesan's congruence theorem relies on it.) The Lenstra–Pomerance algorithm used in the shape test does not use the hyperelliptic method either; it uses the elliptic curve method on "recognisable" smooth numbers [33, §7].

12. Cost assembly and the constant

Proposition 12.1 (cost of one split)**.** *For $N \ge N_0$ as in Section 7, the expected running time of* SPLIT(N) *is at most*

$$L^{\max(a+1/\sqrt{2},\ a+\theta/(2a),\ 2a)+o(1)} = L^{(1+\theta)/\sqrt{2}+o(1)}.$$

Proof. By Lemma 10.2 and Theorem 9.9, the hypotheses of Proposition 7.3(i) hold, so $p \ge L^{-o(1)}$. We bound the remaining quantities in Proposition 7.3(ii).

- *Rows.* $M_{\mathrm{rows}} \le (d+1)\pi(B) + O_d(\log N) = L^{a+o(1)}$.

- *One trial.* SAMPLE_FIELD costs $(\log N)^{O(1)}$ in expectation. DESCEND_ONCE costs one Gauss reduction, two runs of the Lenstra–Pomerance algorithm on integers $\le \sqrt{2N}$, and at most $2k$ special-q searches. The first has expected cost $L_{\sqrt{2N}}[1/2, 1 + o(1)] = L^{1/\sqrt{2}+o(1)}$ (Fact 2.5), uniformly over the inputs. Each search costs $\tilde{O}(B + S^2 \log N) = L^{1/\sqrt{2}+o(1)}$ (Proposition 11.2; recall $S^2 = L^{\theta/(2a)+o(1)}$ and $\theta/(2a) = 1/\sqrt{2}$). AKS certification of the output primes is polynomial. So $C_{\text{trial}} = L^{\max(1/\sqrt{2}, \theta/(2a))+o(1)}$.
- *Post-processing.* $\mathrm{T}_{\text{post}} = M_{\text{rows}}^2 (\log N)^{O(1)} + B^{1+o(1)} = L^{2a+o(1)}$ (Proposition 10.15).

Hence the expected time is at most $L^{o(1)}(L^{a+o(1)} L^{\max(1/\sqrt{2}, \theta/(2a))} + L^{2a})$. With $a = \theta/\sqrt{2}$ the first two exponents both equal $(1 + \theta)/\sqrt{2}$, and $2a = \sqrt{2}\theta < (1 + \theta)/\sqrt{2}$ because $\theta < 1$. □

Remark 12.2 (the choice of a)**.** For a general smoothness exponent a (with $S^2 = L^{\theta/(2a)}$) the same proof gives $L^{c(a)+o(1)}$ with $c(a) = \max(2a, a + 1/\sqrt{2}, a + \theta/(2a))$. For $a > \theta/\sqrt{2}$ the term $a + 1/\sqrt{2}$ exceeds $(1 + \theta)/\sqrt{2}$. On $(0, \theta/\sqrt{2})$ the function $a + \theta/(2a)$ is decreasing, because $\theta/\sqrt{2} < \sqrt{\theta/2}$ for $\theta < 1$. So $a = \theta/\sqrt{2}$ is optimal, and the minimum is $(1 + \theta)/\sqrt{2}$.

13. Uniformity, small factors and prime powers; proof of the main theorem

Proof of Theorem 1.1. Correctness. Every prime in the output is certified by AKS. Every proper factor returned by SPLIT, or by a factor exit, divides the current integer. FACTOR recurses on both cofactors and stops at certified primes. So the output is the complete factorisation, and the algorithm is Las Vegas: it never errs, only its running time is random.

Preprocessing. Step 1 of FACTOR (primality, perfect powers) costs $(\log N)^{O(1)}$. Trial division to $B(N)$ costs $L^{a+o(1)}$. The cofactor N' handed to SPLIT is odd, has no prime factor $\le B(N)$, and is neither a prime nor a perfect power, so $\omega(N') \ge 2$. Since $B(N') \le B(N)$, N' has no prime factor $\le B(N')$, and all statements of Sections 5–12 apply to N' with its own parameters. A non-squarefree N' (for example $p^2 q$) needs no special treatment: Theorem 6.1 uses $|\boldsymbol{\mu}_2| = 2^\omega$, and the other sections use only $\gcd(N', t) = 1$ and $N' \mid F_A(g_0, g_1)$.

Small N'. Step 3 of FACTOR runs SPLIT and trial division in parallel, so its cost is at most twice the smaller of their costs. For $N' < N_0$, trial division bounds it by $O(\sqrt{N_0}) = O(1)$. For $N' \ge N_0$, Proposition 12.1 bounds it by $L_{N'}^{(1+\theta)/\sqrt{2}+o(1)} \le L_N^{(1+\theta)/\sqrt{2}+o(1)}$. Here $N_0 = N_0(d, k, \mu, \eta_0)$ is effective (it involves the effective constants of Facts 10.7–10.9 and of the smooth-number facts), but it need not be known to the algorithm.

Recursion. Every call of SPLIT produces a proper factor, and the factorisation tree has fewer than $\log_2 N$ internal nodes. So there are at most $\log_2 N$ calls of SPLIT and of trial division, each on an integer $\le N$, plus polynomial bookkeeping. The bound of Proposition 12.1 is uniform in the input. To apply it to all smaller inputs at once, replace the $o(1)$ by a non-increasing envelope; this costs only a polylogarithmic factor. Hence the expected total time is $L_N^{(1+\theta)/\sqrt{2}+o(1)}$.

Uniformity. All $L^{o(1)}$ factors are uniform in N: $(\log N)^{O(k)}$, $\binom{2k}{k}$, $2^{O(\omega(N))}$ (Lemma 5.4), $\gamma^{-O(k)}$ (Theorem 9.9), and the Wiedemann and Hensel polylogarithms. No exceptional set of N arises. The analytic inputs need only $\gcd(N, t) = 1$, the absence of prime factors $\le B$, and the coprimality statements of Lemma 3.8. They need no squarefreeness and no condition on the residue classes of the prime factors of N. □

Remark 13.1 (expected time versus probability)**.** Every loop of the algorithm is Las Vegas with verified outputs: primes are certified, kernel vectors checked, $\eta^2 = \Gamma$ tested, and gcds checked. So only expected costs have to be bounded. The Lenstra–Pomerance algorithm is used only through its expected-time bound, so no cap on its running time is needed.

14. A toy implementation

We implemented the whole pipeline at toy scale as a correctness check. It is not evidence for any asymptotic claim. The code is in `src/thmA/` with tests in `tests/test_thmA.py`; the run was pre-registered (commit `9c67cc6`) before it was made.

What is implemented. The pipeline implements the following, and receives only N, a public configuration and a seed.

- The random field with (g_0, g_1) random and $F(g_0, g_1) = Nw$ for a random prime w.
- Dixon descent of uniform squares: Gauss split, shape test, and a special-q atom for every large prime, found with batch smoothness per box.
- The row identity (10) including the split sign.
- The columns of Section 10 with one parity column, first-degree primes $r \nmid f_d \operatorname{disc} f$, and 40 quadratic characters.
- A binary kernel, the Hensel square root in $\mathbb{Z}[\beta]$, the congruence, and the final gcd.

Deviations from the theorem (all disclosed in the pre-registration).

- $d = 3$, $\mu = 1/3$, $H = 4$, and the floors on h are not imposed.
- w is a random prime in $(2M, 4M]$ rather than in $(W, 2W]$. So the algebra of $F(g_0, g_1) = Nw$ is tested, but the analytic role of Lemma 9.3 is not.
- The split shape is relaxed: each half is a smooth part times at most two primes in $(B, 4B]$, with at least one large prime per row. The theorem's shape (at most k primes near $N^{1/(2k)}$) is far too rare at 30–50 bits.
- The shape test uses Pollard–Brent rather than Lenstra–Pomerance, and primality uses deterministic Miller–Rabin rather than AKS.
- The kernel is computed by Gaussian elimination.

Results. Run E1 used 48 instances: 20 balanced semiprimes, 12 three-prime integers, 4 unbalanced, 4 small prime times semiprime, 4 of the form p^2q, and 4 powers or primes, at 30–50 bits. Run E1b used 12 further instances in a diagnostic mode that continues after accidental factors.

registered prediction	result
correct complete factorisation (E1)	48/48 (E1b: 12/12)
row identity (10)	0 violations
congruence $\mathsf{X}_\mathsf{S}^2 \equiv \mathsf{Y}_\mathsf{S}^2$	0 violations
tested kernel vectors that are squares in K	352/352
Theorem 6.1, $\omega = 2$ (predicted 1/2)	$137/288 = 0.476$ ($z = -0.82$)
Theorem 6.1, $\omega = 3$ (predicted 3/4)	$49/64 = 0.766$ ($z = +0.29$)
accidental exits within the null bound	$32 \le 118$
preprocessing (small factors, powers, primes)	12/12

Descriptively, the special-q success rate was 0.06–0.13, and 35–260 fields were tried per kept field. Each split used 83–743 rows and took 0.15–3.1 s. The toy confirms the algebra and the hidden-root symmetry end to end. It cannot test relation supply (Theorems 8.7 and 9.7), which is where the content of the theorem lies, and its running times say nothing about the constant c.

15. Limitations and open problems

- *Not refereed.* The proof was produced and checked with AI assistance only (see the Acknowledgements). An earlier version of Lemma 8.4 imported a bound from inside the proof of Lee–Venkatesan's Lemma 8.5; reviewers named it the weakest link, and it has been replaced by the self-contained proof of Appendix A. The one remaining

external input there, Drappeau's Lemme 5, is published, but its proof refers to an unpublished preprint of Harper. The second moment (Theorem 9.7) is the most novel analytic component, and the field-selection bookkeeping (Proposition 7.3) is the most delicate probabilistic one.

- *Infimum versus a single algorithm.* At fixed d we prove $(1+\theta)/\sqrt{2}+o(1)$ for each fixed k; the infimum over k is not attained. Proving $(1+3/(d+1))/\sqrt{2}+o(1)$ at fixed d would need $k = k(N) \to \infty$ with control of the k-dependence of every constant.
- *The constant* $1/\sqrt{2}$. It comes from factoring two $\sqrt{N}$-size integers with Lenstra–Pomerance. A recursive variant that uses the algorithm itself as its shape test, if it could be made rigorous at all scales, would give the fixed point $c = \sqrt{(1+\sqrt{2})\theta}$ (≈ 0.851 at $d = 9$). We have not proved this.
- *Practicality.* The constants (Stark, Landau, Lagarias–Odlyzko) are effective but astronomically large, and the algorithm is a vehicle for a rigorous bound, not a practical method. The heuristic number field sieve is far faster.
- *Below* $L[1/2]$. This architecture does not reach a rigorous $L[1/3]$. Bounded k forces the $L[1/2, 1/\sqrt{2}]$ shape test, and unbounded k needs averaged estimates of a precision we cannot prove.
- *Other problems.* Whether the same devices (Dixon descent, the random resultant multiplier, the two-dimensional rational side and the family large-sieve bound) improve rigorous bounds for discrete logarithms or class groups is open.

Acknowledgements

This work was carried out with heavy AI assistance. We describe the process so that readers can weigh it.

- *The human author* set the research question and direction and decided which threads to pursue.
- *An orchestrating Claude session* (Anthropic, Claude Opus 5.5) planned the work and reviewed results.
- *Claude Opus 5.5 subagents* produced the proofs, the code, the toy experiments and the project notes on which this paper is based. They worked under written rules (exact arithmetic, no access to the factors in operational code, status labels, and pre-registration of experiments). This paper was drafted by such a subagent from those notes.
- *GPT-6.1-sol* (OpenAI, through the Codex CLI with web search) was used for blind attempts, literature searches and adversarial review. In two places it found the key device independently: the two-dimensional rational randomisation with divisor switching, and the random multiplier of the resultant.

Each component passed at least one GPT-6.1-sol referee and one independent Claude critic. The assembled theorem was audited by two blind GPT-6.1-sol referees and three blind Claude critics, all of whom returned "accept with fixes" with no fatal or major finding. Reviews found and corrected real errors, among them:

- a claimed shape-test cost of $L^{o(1)}$ (actually $L_N[1/2, 1/\sqrt{2}]$), which changed the constant from $\sqrt{2\theta}$ to $(1+\theta)/\sqrt{2}$;
- a second-moment comparison that failed for candidates with small rational value, repaired by the filter $|G(P)| \geq MV/32$;
- a missing sign in the row identity;
- an inference about field selection that ignored factor exits;
- a polylogarithmic lower bound on γ that is false for a computable S;

- the conflation of an infimum over k with an $o(1)$.

No human has checked the proofs line by line. AI review is not peer review.

Appendix A. Proof of Lemma 8.4

This appendix gives a self-contained proof of Lemma 8.4. It does not use Lee–Venkatesan's Lemma 8.5, neither its statement nor its proof. It also does not use Harper's preprint [18] directly.

- For moduli $t = \ell s$ with ℓ a prime $> B$, the part of the error coming from characters principal at ℓ is *exactly* the error modulo s divided by $\ell - 1$ (Lemma A.8). So we need a restricted Bombieri–Vinogradov estimate over smooth moduli with a relative saving, which we prove as Lemma A.16.
- Characters of large conductor are handled by the classical multiplicative large sieve and a Perron bound (Lemma A.14).
- Characters of small conductor are handled by one published lemma of Drappeau (Fact A.2).

The proof follows the project note `notes/lem85_results.md`, which was checked by one GPT-6.1-sol and two Claude referees.

Honest statement of the one external input. Fact A.2 is a published, numbered statement: Drappeau [14, Lemme 5], restated by Pascadi [39, Lemma 3.2]. In both sources, however, its proof consists of a reference to the computations of §3.3 of Harper's preprint [18], which has not been published. Drappeau writes "il découle des calculs de la section 3.3 de [Har12b]". Our argument never enters the preprint, but the preprint stands behind that published statement. Removing it would mean re-proving Harper's zero-density and zero-free-region analysis of $\Psi(x, y; \chi)$, with Page's theorem, and we have not done that. We apply Fact A.2 with $A = 1/2$ and discard the factor $(\log x)^{-A}$. So Siegel's theorem is never needed, and only the effective part of Harper's computation is involved.

A.1. **Setting and statement.** Throughout, $y := B = \lfloor L^a \rfloor$, and $S(z, y)$ is the set of y-smooth $n \le z$. We write $\Psi(z, y) = \#S(z, y)$, $\varrho(z, y) = \Psi(z, y)/z$ and $\Psi(z, y; \chi) = \sum_{n \in S(z,y)} \chi(n)$. For $(a, q) = 1$ put $E(z, y; q, a) := \Psi(z, y; q, a) - \Psi_q(z, y)/\varphi(q)$, and let $\Delta(q) := \varphi(q)^{-1} \sum_{\chi \neq \chi_0 \bmod q} |\Psi(z, y; \chi)|$. Let $H(u) := \exp(u/\log^2(u + 1))$ and $u = \log x/\log y$. Let $\alpha(z, y)$ be the Hildebrand–Tenenbaum saddle point. $P^+(n)$ and $P^-(n)$ denote the largest and smallest prime factors.

Regime (R). Fix $a > 0$, a compact $\Xi \subset (0, \infty)$ with $\xi_{\max} := \max \Xi$, and $\lambda \in (0, 1/4)$. Let x be real with $\log x/\log N \in \Xi$. Put $K := 6 + \lceil 2\xi_{\max}/a^2 \rceil$, $R_1 := y^K$, and $\varepsilon_N := \exp\bigl(-c_0(\log N)^{1/2}/(\log\log N)^{5/2}\bigr)$, where $c_0 := \delta_D \min \Xi/(2a)$ and δ_D is the constant of Fact A.2.

Lemma A.1 (general form of Lemma 8.4)**.** *In regime (R), let Y, Q satisfy $y^K \le Y$ and $x^\lambda Y \le Q \le x^{1/2-\lambda}$. Put $J_Y := \{\ell \text{ prime} : Y < \ell \le 2Y\}$, $\mathcal{T}_Q := \{t = \ell s \in (Q/2, Q] : \ell \in J_Y,\ P^+(s) \le y\}$ and $\delta_{\mathcal{T}} := \varrho(Q/Y, y)/\log Y$. Then there are N_0, C, depending only on (a, Ξ, λ) and on the constants of the cited results, such that for $N \ge N_0$:*

(i) $Q\delta_{\mathcal{T}}/10 \le \#\mathcal{T}_Q \le 3Q\delta_{\mathcal{T}}$;

(ii) $\sum_{t \in \mathcal{T}_Q} \max_{(r,t)=1} |E(x, y; t, r)| \le C\, \Psi(x, y)\, \delta_{\mathcal{T}}\, \varepsilon_N$.

Deduction of Lemma 8.4. In Lemma 8.4, $Y = N^{2\nu+o(1)} \ge y^K$ and $Q/Y = N^{q_0-2\nu+o(1)}$. Take Ξ a small closed neighbourhood of ξ and $\lambda := \min(q_0 - 2\nu,\ \xi/2 - q_0)/(4\xi)$. Then $x^\lambda Y \le Q \le x^{1/2-\lambda}$. By (i), $\delta_{\mathcal{T}} \le 10\#\mathcal{T}_Q/Q$, so (ii) gives Lemma 8.4 with this ε_N. In the regime of Lemma 8.4, $u \asymp (\log N/\log\log N)^{1/2}$, so ε_N beats every power of $\log N$. The bound is uniform over the grid $\mathcal{X}$ and the $O(1)$ dyadic values of Q used in Theorem 8.7. □

A.2. **Cited results.**

Fact A.2 (Drappeau [14, Lemme 5]; Pascadi [39, Lemma 3.2])**.** *There exist constants $\eta_D, c_D, \delta_D > 0$ such that, whenever $(\log x)^{c_D} \le y \le x$ and $Q \le x$, for every fixed $A > 0$,*

$$\sum_{q \le Q} \frac{1}{\varphi(q)} \sum_{\substack{\chi \bmod q \\ 1 < \mathrm{cond}(\chi) \le x^{\eta_D}}} \Big| \sum_{n \in S(x,y)} \chi(n) \Big| \ll_A \Psi(x,y)\big\{H(u)^{-\delta_D}(\log x)^{-A} + y^{-\delta_D}\big\}.$$

The implied constant is effective if $A < 1$.

Drappeau states effectivity for the implied constant when $A < 1$, but not for η_D, c_D, δ_D. Harper fixes these explicitly from effective inputs, so we believe they are effective, but no published source says so. Our use lies within Harper's literal setting: we take $Q = R_1 = y^K \le \sqrt{\Psi}$ and discard $(\log x)^{-A}$. The algorithm never uses these constants (see Remark A.17).

Fact A.3 (Drappeau [14, Lemme 1(i)], from de la Bretèche–Tenenbaum [12, Thm. 2.4(i)])**.** *For $2 \le y \le x$ and $1 \le d \le x$, $\Psi(x/d, y) \le C_{\mathrm{dBT}} \Psi(x,y) d^{-\alpha(x,y)}$, with C_{dBT} absolute.*

Fact A.4 (Iwaniec–Kowalski [24, Thm. 7.13])**.** *For complex a_n supported on $M < n \le M + N'$ and integers $Q \ge 1$, $\sum_{q \le Q} \frac{q}{\varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} \big|\sum_n a_n \chi(n)\big|^2 \le (Q^2 + N' - 1) \sum |a_n|^2$, where $\sum^*$ runs over primitive characters.*

Fact A.5 (Rosser–Schoenfeld [45])**.** *$\pi(z) < 1.25506 z/\log z$ for $z > 1$; $\pi(2z) - \pi(z) > 3z/(5 \log z)$ for $z \ge 20.5$; and $n/\varphi(n) < e^\gamma \log\log n + 2.51/\log\log n$ for $n \ge 3$.*

Hildebrand–Tenenbaum (Fact 2.2) gives, in regime (R) and for $N \ge N_0$:

(α1) $3/4 \le \alpha(z,y) \le 1$ for $2y \le z \le x$;
(α2) $1 - \alpha(z,y) \le (\log(u \log y) + C_\alpha \log\log y)/\log y$ for $2y \le z \le x$.

To see (α2), note that $\log(1 + y/\log z) \ge \log y - \log\log z$ and $\log z \le u \log y$. (α1) follows in regime (R).

Lemma A.6 (truncated Perron kernel)**.** *For $c > 0$, $T > 0$ and $w > 0$ with $w \ne 1$, $\big|\frac{1}{2\pi i} \int_{c-iT}^{c+iT} w^s \frac{ds}{s} - 1_{w>1}\big| \le \frac{w^c}{\pi T |\log w|}$.*

Proof. The integrand w^s/s has one simple pole, at $s = 0$, with residue 1. If $w > 1$, close the contour with a rectangle whose left side is at $\Re s = -\sigma_0$, $\sigma_0 \to \infty$. The left side contributes at most $w^{-\sigma_0} 2T/\sigma_0 \to 0$, and each horizontal side at most $(2\pi)^{-1} \int_{-\infty}^{c} w^\sigma d\sigma/T = w^c/(2\pi T \log w)$. If $w < 1$, close to the right; there is no pole. □

A.3. **Elementary lemmas.**

Lemma A.7 (orthogonality)**.** *For $(a,q) = 1$, $E(z,y;q,a) = \varphi(q)^{-1} \sum_{\chi \ne \chi_0} \bar\chi(a) \Psi(z,y;\chi)$. Hence $\max_{(a,q)=1} |E(z,y;q,a)| \le \Delta(q)$.*

Proof. Orthogonality of characters, together with $\Psi(z,y;\chi_0) = \Psi_q(z,y)$. □

Lemma A.8 (lift averaging)**.** *Let $\ell > y$ be prime, let $\ell \nmid s$, and put $t = \ell s$. For every $(r,t) = 1$,*

$$E(z,y;t,r) = \frac{E(z,y;s,r)}{\ell - 1} + \mathrm{Rem}(t,r), \qquad |\mathrm{Rem}(t,r)| \le \Delta_\ell(t) := \frac{1}{\varphi(t)} \sum_{\chi \bmod t,\ \ell \mid \mathrm{cond}\chi} |\Psi(z,y;\chi)|.$$

Proof. Every character modulo t is $\chi_\ell \chi_s$, and $\ell \mid \mathrm{cond}\chi$ iff χ_ℓ is non-principal. Every y-smooth n is coprime to ℓ, so $\Psi(z,y;\chi_0^{(\ell)}\chi_s) = \Psi(z,y;\chi_s)$. Also $\Psi_t = \Psi_s$ and $\varphi(t) = (\ell - 1)\varphi(s)$. In Lemma A.7 for the modulus t, the terms with χ_ℓ principal and χ_s non-principal add up to $E(z,y;s,r)/(\ell - 1)$. □

Lemma A.9 (coprimality by Möbius)**.** *Let $\chi \bmod q$ be induced by a primitive $\chi^* \bmod r$, and put $q_y := \prod_{p\mid q,\, p\le y} p$. Then $\Psi(z,y;\chi) = \sum_{d\mid q_y,\,(d,r)=1} \mu(d)\chi^*(d)\Psi(z/d,y;\chi^*)$.*

Proof. For y-smooth n, $(n,q) = 1$ iff $(n,q_y) = 1$. Expand $1_{(n,q_y)=1} = \sum_{d\mid(n,q_y)} \mu(d)$ and write $n = dn'$ with $n' \in S(z/d,y)$. Terms with $(d,r) > 1$ vanish. □

Lemma A.10 (factorisation)**.** *Let $M_1 \ge 1$. Every $n \in S(z,y)$ with $n > M_1$ has exactly one factorisation $n = mk$ with $M_1 < m \le P^+(m)M_1$, $P^+(m) \le P^-(k)$ and $k \in S(z,y)$. Moreover $k \le z/M_1$.*

Proof. List the primes of n in increasing order and let m be the shortest initial product exceeding M_1. Conversely, the two conditions force the primes of m to come first, and $m/P^+(m) \le M_1 < m$ fixes their number. □

Lemma A.11 (constants)**.** *(i) $\sum_{j\le X} 1/\varphi(j) \le 2(1+\log X)$ for $X \ge 1$.*

(ii) Over squarefree d: $\sum_d 1/(d\varphi(d)) < 1.95$, $\sum_d 1/(\varphi(d)d^{1/2}) < 6$, and $\sum_d d^{1-\beta-\beta'}/\varphi(d) < 6$ if $\beta,\beta' \ge 3/4$.

(iii) $n/\varphi(n) \le 5\log\log(n+16)$ for all $n \ge 1$.

Proof. (i) $j/\varphi(j) = \sum_{e\mid j} \mu^2(e)/\varphi(e)$, so the sum is at most $(1+\log X)\sum_e \mu^2(e)/(e\varphi(e))$, and $\sum_e \mu^2(e)/(e\varphi(e)) = \zeta(2)\zeta(3)/\zeta(6) < 1.95$. (ii) The Euler products are $\prod_p(1+2p^{-3/2}) \le \exp(2\sum_p p^{-3/2}) < e^{1.7} < 6$ and $\prod_p(1+p^{-1/2}/(p-1)) < 6$. (iii) Fact A.5 for $n \ge 16$, and inspection for $n < 16$. □

Lemma A.12 (density lower bound)**.** *For $2y \le z \le x$ and $N \ge N_0$, $\varrho(z,y) \ge y^{-1}\exp(-u\log(4u\log y))$. In regime (R), $\varrho(z,y) \ge y^{-1-\xi_{\max}/a^2-o(1)}$.*

Proof. Let $k_0 := \lfloor \log z/\log y \rfloor \le u$, and let $\mathcal{P}$ be the set of primes in $(y/2,y]$, so $\#\mathcal{P} \ge 3y/(10\log y) \ge k_0$ by Fact A.5. Products of k_0 distinct primes of $\mathcal{P}$ are distinct, y-smooth and at most $y^{k_0} \le z$. So $\Psi(z,y) \ge \binom{\#\mathcal{P}}{k_0} \ge (z/y)(3/(10k_0\log y))^{k_0}$. In regime (R), $u\log(4u\log y) = (\xi/a^2)(1+o(1))\log y$. □

Lemma A.13 (weights over smooth moduli)**.** *Let $2y \le Q' \le x$, let $r,d \ge 1$ with $(r,d) = 1$, and put $\beta' := \alpha(Q',y)$. Then*

$$\sum_{\substack{Q'/2 < s \le Q',\ P^+(s)\le y \\ rd\mid s}} \frac{1}{\varphi(s)} \;\le\; 10C_{\mathrm{dBT}}\log\log(Q'+16)\,\varrho(Q',y)\frac{(rd)^{1-\beta'}}{\varphi(r)\varphi(d)}.$$

Proof. Write $s = rdj$, so $1/\varphi(s) \le 1/(\varphi(r)\varphi(d)\varphi(j))$, with j y-smooth in $(J'/2,J']$, $J' := Q'/(rd)$. By Lemma A.11(iii), $\sum_j 1/\varphi(j) \le 5\log\log(Q'+16)(2/J')\Psi(J',y)$. By Fact A.3, $\varrho(J',y) = \Psi(Q'/(rd),y)rd/Q' \le C_{\mathrm{dBT}}\varrho(Q',y)(rd)^{1-\beta'}$. □

A.4. **Large conductors.**

Lemma A.14 (large conductors)**.** *Let $y \ge 2$, $z \ge \max(4,y^{3/2})$ and $R \ge 1$. Then*

$$\sum_{R<r\le 2R} \frac{r}{\varphi(r)} \sideset{}{^*}\sum_{\chi \bmod r} |\Psi(z,y;\chi)| \;\le\; 30\pi(y)\log(2zR)\big(y^{3/2}z + y^{1/2}z^{1/2}R^2\big).$$

Proof. Write $w_\chi := r/\varphi(r)$, so $\sum_\chi w_\chi \le 4R^2$. Put $M_1 := z^{1/2}$. By Lemma A.10,

$$\Psi(z,y;\chi) = \sum_{n\le M_1,\, n\in S(z,y)} \chi(n) + \sum_{p\le y}\sum_{mk\le z} a_m^{(p)} b_k^{(p)} \chi(m)\chi(k),$$

with $a_m^{(p)} := 1$ if $P^+(m) = p$ and $M_1 < m \le pM_1$, and $b_k^{(p)} := 1$ if $P^-(k) \ge p$, $P^+(k) \le y$, $k \le z/M_1$ (and 0 otherwise). Neither sequence depends on χ, and their supports have sizes at most yM_1 and z/M_1.

Short part. Cauchy–Schwarz and Fact A.4 with $Q = \lfloor 2R \rfloor$ give at most $(4R^2)^{1/2}((4R^2 + M_1)M_1)^{1/2} \le 2Rz^{1/2} + 4R^2 z^{1/4}$.

Bilinear part. Let $z' := \lfloor z \rfloor + \frac{1}{2}$, $c := 1/\log(2z)$ and $T := (2zR)^4$. For integers $m, k \ge 1$ we have $1_{mk \le z} = 1_{z'/(mk) > 1}$. On the support, $mk \le yz$ and $|z' - mk| \ge 1/2$, so $|\log(z'/mk)| \ge 1/(4yz)$ and $(z'/mk)^c \le e$. By Lemma A.6 each pair contributes an error at most $4eyz/(\pi T)$. Summing over at most yz pairs per p, at most y primes p, and total weight $4R^2$, the error is at most $16ey^3z^2R^2/T \le 3$. The main term is $(2\pi i)^{-1} \int_{c-iT}^{c+iT} A_p(\chi, s)B_p(\chi, s)z'^s \, ds/s$, with $A_p(\chi, s) = \sum_m a_m^{(p)} \chi(m) m^{-s}$ and B_p similarly. For each t, Cauchy–Schwarz and Fact A.4 give

$$\sum_\chi w_\chi |A_p B_p|(c+it) \le (yM_1 + 4R^2)^{1/2}(z/M_1 + 4R^2)^{1/2}(yM_1)^{1/2}(z/M_1)^{1/2} \le y^{3/2}z + 4y^{1/2}z^{1/2}R^2.$$

Also $\int_{-T}^{T} dt/|c + it| \le 2/c + 2\log T \le 10\log(2zR)$ and $z'^c \le e$. Summing over $p \le y$ and adding the short part gives the bound. □

Corollary A.15 (moduli with smooth cofactor)**.** *Let $z_0 \ge \max(4, y^{3/2})$ and $1 \le R_0 \le Q_0$ with $z_0/Q_0 \ge \max(4, y^{3/2})$. For each $q \le Q_0$ let $\mathcal{X}(q)$ be any set of characters modulo q whose conductors lie in $(R_0, Q_0]$. Then*

$$\sum_{q \le Q_0} \frac{1}{\varphi(q)} \sum_{\chi \in \mathcal{X}(q)} |\Psi(z_0, y; \chi)| \;\le\; 1.5 \cdot 10^3 \pi(y)(\log z_0)^2 \big(y^{3/2} z_0/R_0 + y^{1/2} z_0^{1/2} Q_0\big).$$

Proof. Group (q, χ) by the inducing primitive $\chi^* \bmod r$ and apply Lemma A.9: $|\Psi(z_0, y; \chi)| \le \sum_{d | q_y, (d,r)=1} |\Psi(z_0/d, y; \chi^*)|$, with d squarefree and $d \le Q_0$. By Lemma A.11(i) and $\varphi(rdj) \ge \varphi(r)\varphi(d)\varphi(j)$, we have $\sum_{q \le Q_0,\, rd | q} 1/\varphi(q) \le 2(1 + \log Q_0)/(\varphi(r)\varphi(d))$. On each dyadic block $r \in (R', 2R']$ with $R_0 \le R' \le Q_0$, use $1/\varphi(r) \le (r/\varphi(r))/R'$ and Lemma A.14 at $z = z_0/d \ge z_0/Q_0$. Here $\log(2zR') \le 2\log z_0$. Sum over the blocks ($\sum 1/R' \le 2/R_0$, $\sum R' \le 2Q_0$) and then over d with Lemma A.11(ii). Finally $1 + \log Q_0 \le \log z_0$. □

A.5. **Smooth moduli with a relative saving.**

Lemma A.16 (smooth moduli)**.** *In regime (R), let $2y \le Q' \le x^{1/2-\lambda}$ and $\mathcal{S}' := \{s \in (Q'/2, Q'] : P^+(s) \le y\}$. Then for $N \ge N_0$,*

$$\sum_{s \in \mathcal{S}'} \max_{(a,s)=1} |E(x, y; s, a)| \le \sum_{s \in \mathcal{S}'} \Delta(s) \le C_{\mathrm{SM}} \Psi(x, y) \varrho(Q', y) \varepsilon_N.$$

Proof. The first inequality is Lemma A.7. Split $\Delta(s)$ by the conductor r of χ: case (a) $1 < r \le R_1$, case (b) $r > R_1$.

(a) Small conductors. Apply Lemma A.9 (s is y-smooth) and exchange the sums. Only $d \le Q'$ occurs. By Lemma A.13 with $\beta' = \alpha(Q', y)$,

$$(a) \le 10 C_{\mathrm{dBT}} \log\log(Q' + 16) \varrho(Q', y) R_1^{1-\beta'} \sum_{d \le Q'} \frac{d^{1-\beta'}}{\varphi(d)} \mathsf{D}(x/d),$$

$$\mathsf{D}(z) := \sum_{1 < r \le R_1} \frac{1}{\varphi(r)} \sideset{}{^*}\sum_{\chi \bmod r} |\Psi(z, y; \chi)|,$$

with d squarefree, y-smooth and coprime to r. Apply Fact A.2 to the primitive characters only (the terms $q = r$ of its left side, with $Q := R_1$), with $A = 1/2$ and argument x/d.

Its hypotheses hold: $x/d \ge x^{1/2}$, so $(\log(x/d))^{c_D} \le y$, and $R_1 = N^{o(1)} \le (x/d)^{\eta_D}$. Hence $\mathsf{D}(x/d) \le C_D\Psi(x/d, y)(H(u/2)^{-\delta_D} + y^{-\delta_D})$, because $u_d := \log(x/d)/\log y \ge u/2$ and H is increasing on $[4, \infty)$. By Fact A.3, $\Psi(x/d, y) \le C_{\mathrm{dBT}}\Psi(x, y)d^{-\beta}$ with $\beta = \alpha(x, y)$. By (α1) and Lemma A.11(ii), $\sum_d d^{1-\beta-\beta'}/\varphi(d) < 6$. By ($\alpha$2), $R_1^{1-\beta'} \le (\log x)^{K(1+C_\alpha)}$. Therefore

$$(a) \le 60C_{\mathrm{dBT}}^2 C_D \log\log x\,(\log x)^{K(1+C_\alpha)}\big(e^{-\delta_D(u/2)/\log^2(u/2+1)} + y^{-\delta_D}\big)\Psi(x, y)\varrho(Q', y).$$

In regime (R), $\delta_D(u/2)/\log^2(u/2+1) = (2\delta_D\xi/a)(\log N)^{1/2}(\log\log N)^{-5/2}(1+o(1))$. This beats $(\log x)^{O(1)}$ and gives $(a) \le \Psi(x, y)\varrho(Q', y)\varepsilon_N$.

(b) Large conductors. If $Q' \le R_1$, case (b) is empty. Otherwise apply Corollary A.15 with $z_0 = x$, $Q_0 = Q'$, $R_0 = R_1$, keeping only the moduli in $\mathcal{S}'$. This gives $(b) \le 1.5 \cdot 10^3\pi(y)(\log x)^2(y^{3/2}x/y^K + y^{1/2}x^{1/2}Q')$. By Lemma A.12, $\Psi(x, y)\varrho(Q', y) \ge xy^{-2-2\xi_{\max}/a^2-o(1)}$. The first term is then at most $\Psi\varrho(Q')y^{-1}$ by the choice of K, and the second at most $\Psi\varrho(Q')x^{-\lambda}y^{O(1)}$, since $Q' \le x^{1/2-\lambda}$. □

The argument works exactly at $y = L^{\Theta(1)}$. The large sieve needs $\log R_1 \gtrsim u\log u$, while the small-conductor transfer costs $R_1^{1-\beta'} = \exp(\log R_1 \cdot \log(u\log y)/\log y)$. Both are compatible with $\log R_1 \asymp \log y$ precisely when $u\log u \asymp \log y$.

A.6. **Proof of Lemma A.1.** *(i).* Since $\ell > y \ge P^+(s)$, each t determines (ℓ, s), so $\#\mathcal{T}_Q = \sum_{\ell\in J_Y}(\Psi(Q/\ell, y) - \Psi(Q/2\ell, y))$. The upper bound follows from $\#J_Y \le 1.256 \cdot 2Y/\log 2Y$ and $\Psi(Q/\ell, y) \le (Q/Y)\varrho(Q/Y, y)$. For the lower bound put $z := Q/(2\ell) \ge x^\lambda/4$. Then Fact 2.2 with $c = 2$ and $c = 4$ (and $\alpha \to 1$) gives $\Psi(2z, y) - \Psi(z, y) \ge 0.8\Psi(z, y)$ and $\Psi(Q/Y, y) \le 4.1\Psi(z, y)$. So each ℓ contributes at least $0.19(Q/Y)\varrho(Q/Y, y)$, and $\#J_Y \ge 3Y/(5\log Y)$.

(ii). By Lemma A.8, $\max_{(r,t)=1}|E(x, y; t, r)| \le \max_{(a,s)=1}|E(x, y; s, a)|/(\ell - 1) + \Delta_\ell(t)$.

Part 1 (characters principal at ℓ). The sum of the first terms is

$$\sum_{\ell\in J_Y}\frac{1}{\ell - 1}\sum_{s\in(Q/2\ell, Q/\ell],\, P^+(s)\le y}\max_a|E(x, y; s, a)|.$$

Lemma A.16 applies with $Q' = Q/\ell$, since $Q/\ell \ge x^\lambda/2 \ge 2y$ and $Q/\ell \le x^{1/2-\lambda}$. It gives at most $C_{\mathrm{SM}}\Psi(x, y)\varepsilon_N\sum_\ell \varrho(Q/\ell, y)/(\ell - 1)$. Now $\varrho(Q/\ell, y) \le 2\varrho(Q/Y, y)$ and $\sum_{\ell\in J_Y} 1/(\ell - 1) \le 2.6/\log Y$, so Part 1 is at most $5.2C_{\mathrm{SM}}\Psi(x, y)\delta_{\mathcal{T}}\varepsilon_N$.

Part 2 (characters with $\ell \mid$ conductor). Every such conductor exceeds Y. Apply Corollary A.15 with $z_0 = x$, $Q_0 = Q$, $R_0 = Y$, and $\mathcal{X}(t) = \{\chi \bmod t : \ell_t \mid \mathrm{cond}\chi\}$ for $t \in \mathcal{T}_Q$ (empty otherwise). This gives at most $1.5 \cdot 10^3\pi(y)(\log x)^2(y^{3/2}x/Y + y^{1/2}x^{1/2}Q)$. By Lemma A.12, $\Psi(x, y)\delta_{\mathcal{T}} \ge xy^{-2-2\xi_{\max}/a^2-o(1)}/\log Y$. Since $Y \ge y^K$ and $Q \le x^{1/2-\lambda}$, Part 2 is at most $\Psi(x, y)\delta_{\mathcal{T}}y^{-1/2} \le \Psi(x, y)\delta_{\mathcal{T}}\varepsilon_N$.

Adding the two parts proves (ii). □

Remark A.17 (exceptional zeros, effectivity, and the algorithm)**.** Only case (a) of Lemma A.16 could see a Siegel or Page zero, and it is handled entirely through Fact A.2 with $A = 1/2$. Every other step uses no information about zeros of L-functions. There is no exceptional set of N and no exceptional set of moduli. The algorithm uses none of the analytic constants: its parameters B, S, M, H, W, J and $\mathcal{X}$ are explicit functions of N and of (d, k, μ, η_0). The only place where a threshold N_0 could enter the algorithm is the small-N branch, and FACTOR removes it by alternating trial division with SPLIT step by step (Algorithm 3.10, step 3). This costs at most a factor 2 in expected time. So even if some constant behind Fact A.2 were ineffective, the algorithm would be explicit and Theorem 1.1 would be unaffected; only the threshold beyond which its time bound applies would become ineffective.

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